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Field-Circuit Co-simulation of a Planar Schottky Diode

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Abstract

This project implements and validates a drift-diffusion model (DDM) for the simulation of the planar Gallium Arsenide (GaAs) Schottky diodes. The nonlinear carrier transport in the diodes is described by the coupled Poisson and the continuity equations, which are solved using the continuous Galerkin finite element method (FEM) with successive under-relaxation. The developed solver enables detailed investigations of carrier distributions, potential profiles and current-voltage $(I - V)$ characteristics. Furthermore, the simulated results from the DDM show good agreement with those obtained from the commercial software COMSOL. Compared to the traditional lumped-element model, the solver offers a physically consistent and computationally efficient tool for analyzing Schottky diode behavior, particularly relevant for high frequencies (THz) applications.

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1 Introduction

In communication electronics, nonlinear elements such as semiconductor diodes are crucial for frequency conversions. They can be used as a mixer, up- and down-mixing the information carrying modulated signals, achieving wireless transmission [1]. Among them, planar GaAs Schottky diodes are widely employed in high frequency studies, particularly in the terahertz regime [2]. Figure 1 shows the diode structure of interest along with its doping profile investigated in this study.

As operating frequencies increase and device dimensions shrink, the traditional, widely-used lumped-element models fail to capture [3] the distributed nature of the carriers or the field interaction within the semiconductor structure, leading to inaccurate modeling [4].

To overcome these limitations, the physics-based drift-diffusion model (DDM) [5] aims at accurately representing the semiconductor model, by solving nonlinear carrier transport equations, under drift and diffusion schemes. And then it is coupled with external voltage sources, to output electric characteristics within the structure, building foundation for circuit coupling [6]. In this project, DDM is implemented with the finite element method (FEM), like in other commercial tools such as COMSOL [7], which is also used to verify our final simulation results. They show good agreement and confirm the accuracy of the implementation. The primary contribution of this work is thus a transparent and extendable modeling framework for Schottky diodes at the device level.

The next phase of this work will focus on two key extensions. First, the developed modeling framework must be expanded to support both DC and AC analysis to fully characterize the diode's static and dynamic high-frequency operation. Second, and crucially, the validated semiconductor model will be integrated with external components by implementing the necessary algorithms for field-circuit coupling, enabling the co-simulation of device within a simple test circuit.

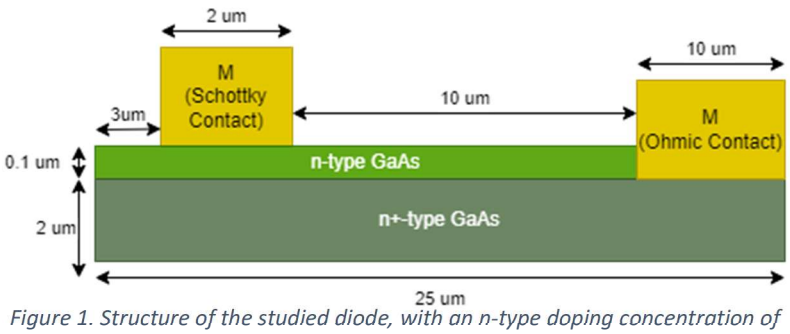


Figure 1. Structure of the studied diode, with an n-type doping concentration of $2 \cdot 10^{17} cm^{-3}$ *and n⁺-type doping concentration of* $2 \cdot 10^{18} cm^{-3}$.

2 Theory

This chapter introduces the theoretical background for the modeling and simulation of the planar GaAs Schottky diodes. It consists of the essential physical and mathematical framework, such as the semiconductor transport theory with the Schottky and Ohmic contacts, DDM and FEM formulations.

2.1 Semiconductor Fundamentals

The behavior of carriers in a semiconductor device is governed by both electrostatics and carrier transport. Specifically, the electrostatic potential within the device is described by Poisson's equation [8]:

$$\nabla \cdot (\varepsilon \nabla \varphi) = -q(p - n + N_D^+ - N_A^-), \tag{2.1}$$

where ε is the dielectric permittivity, φ is the electrostatic potential, n and p are the electron and hole concentrations, respectively, and N_D^+ and N_A^- are the ionized donor and acceptor concentrations, respectively. Carrier transport is modeled by the drift-diffusion equations [8], which describe electron and hole current densities as

$$\boldsymbol{J}_n = qn\mu_n(-\nabla\varphi) + qD_n\nabla n, \tag{2.2}$$

$$\boldsymbol{J}_p = qp\mu_p(-\nabla\varphi) - qD_p\nabla p, \tag{2.3}$$

with μ_n, μ_p being the mobilities, and D_n, D_p the diffusion coefficients of electrons and holes, respectively. These quantities are related via the Einstein relation [8]:

$$D_n = \frac{k_B T}{q}\mu_n, \tag{2.4}$$

$$D_p = \frac{k_B T}{q}\mu_p\,. \tag{2.5}$$

Charge conservation is enforced by the continuity equations [8]:

$$\frac{\partial n}{\partial t} = \frac{1}{q}\nabla \cdot \boldsymbol{J}_n + G - R, \tag{2.6}$$

$$\frac{\partial n}{\partial t} = \frac{1}{q}\nabla \cdot \boldsymbol{J}_n + G - R, \tag{2.7}$$

where G and R are the generation and recombination rates, respectively.

Altogether, the Poisson, the drift-diffusion, and the continuity equations build up DDM. This model provides the physical basis for describing the nonlinear transport behavior of the Schottky diodes, achieving the device-level semiconductor simulation.

2.2 Metal-Semiconductor Contact

As shown in the device structure under study in Figure 1, the planar GaAs diode comprises two electrodes that form different types of metal-semiconductor (MS) contacts. On the left-hand-side and top of the semiconductor, the metal forms a Schottky contact with the n-type GaAs layer. On the right-hand-side and top of the heavily doped n^+-GaAs, an Ohmic contact is formed. These two contacts govern the injection and extraction of carriers and thus define the boundary conditions for device simulation.

To better understand how the contacts behave, the band alignment between a metal and a

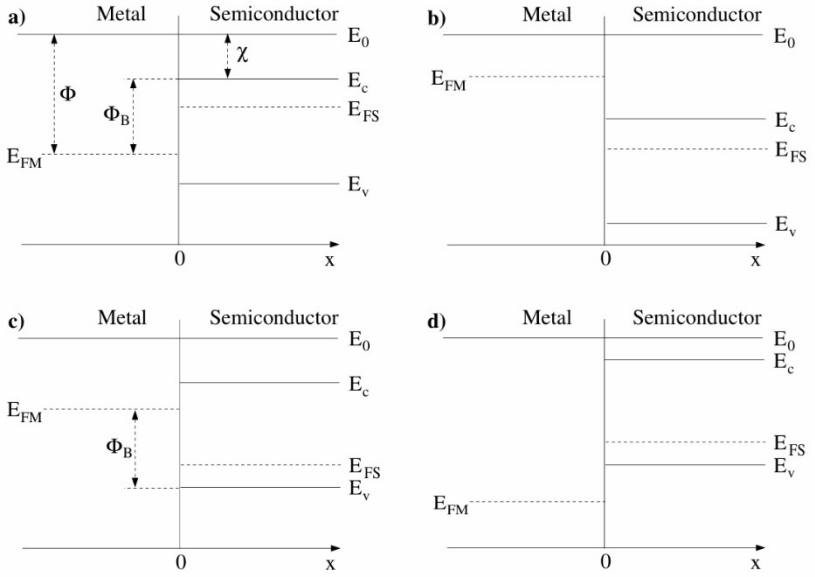


Figure 2. Energy band diagrams of MS contacts immediately after formation, before reaching thermodynamic equilibrium. Case a): n-type semiconductor with $E_{FS} > E_{FM}$, case b): n-type semiconductor with $E_{FS} < E_{FM}$, case c): p-type semiconductor with $E_{FS} < E_{FM}$, case d): p-type semiconductor with $E_{FS} > E_{FM}$.

semiconductor at the moment of contact formation is illustrated in Figure 2 [9].

The relative positions of the metal Fermi level E_{FM} and the semiconductor Fermi level E_{FS}, together with the electron affinity χ and the metal work function ϕ, determines whether the contact forms a barrier or allows nearly ideal carrier transport.

- Case a): For n-type semiconductors, if the metal work function is significantly higher than the semiconductor affinity, a Schottky barrier ϕ_B forms, which impedes electron flow from semiconductor to metal and gives rise to rectifying behavior.
- Case b): Also, for n-type semiconductors, if the metal work function is low enough, the barrier vanishes or becomes very small, approaching Ohmic-like conduction.
- Case c): For p-type semiconductors, if the metal work function is too low, a Schottky barrier appears for holes, preventing hole injection into the metal and again resulting in rectifying behavior.
- Case d): For p-type semiconductors, when the metal work function is sufficiently high, the barrier for holes becomes very small or disappears, and the contact tends to be Ohmic.

2.2.1 Schottky Contact Theory

When a metal is brought into contact with a semiconductor, the difference between the metal work function ϕ and the semiconductor electron affinity χ determines the barrier height at the interface. For an n-type semiconductor, the Schottky barrier height ϕ_B is expressed as

$$\phi_B = \phi - \chi. \tag{2.8}$$

After the thermodynamic equilibrium is reached, band bending occurs near the interface, as shown in Figure 3 (a) and (c), for n-type and p-type semiconductors, respectively. In the n-type case, electrons in the conduction band face a potential barrier ϕ_B, restricting their flow into the metal. This barrier is the origin of the rectifying behavior characteristic of Schottky contacts.

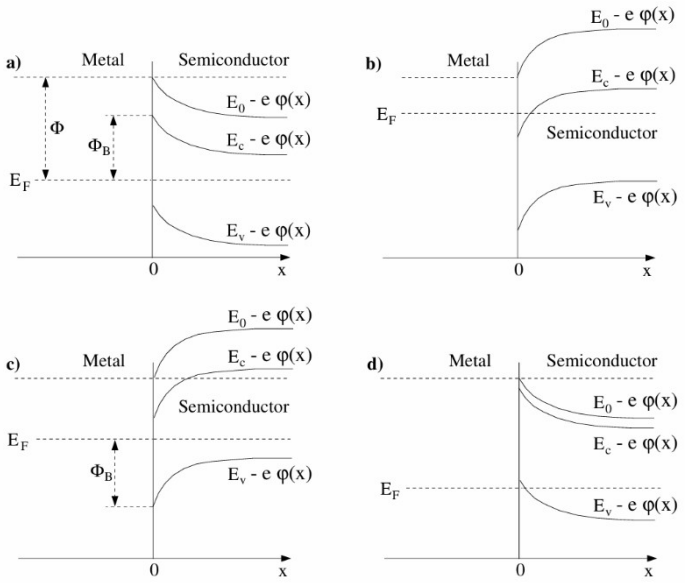


Figure 3. Potential dependence of the energy levels of the MS junction after thermodynamic equilibrium is established. Cases a) and c) are called Schottky contact, cases b) and d) Ohmic contacts.

Carrier transport across the Schottky barrier is primarily governed by thermionic emission theory [10], which gives the current density as

$$J = A^* T^2 \exp\left(-\frac{q\phi_B}{k_B T}\right)\left[\exp\left(\frac{qV}{k_B T}\right) - 1\right], \tag{2.9}$$

where A^* is the Richardson constant, T is the absolute temperature, V is the applied bias, q is the elementary charge, and k_B is Boltzmann's constant, directly reflecting the $I - V$ characteristics.

In the numerical implementation, the externally applied bias V_0 cannot be directly imposed on the semiconductor potential at the Schottky boundary. Instead, it must be converted into an effective potential that accounts for the Schottky barrier height and the electron affinity of the semiconductor:

$$V = V_0 - \phi_B - \chi - V_{eq,adj}, \tag{2.10}$$

where $V_{eq,adj}$ is the equilibrium adjustment term to align with the Fermi levels at the interface. Moreover, the normal components of the electron and hole currents at the Schottky boundary are given by

$$\boldsymbol{J}_n \cdot \boldsymbol{n} = -qv_{s,n}(n - n_0), \tag{2.11}$$

$$\boldsymbol{J}_p \cdot \boldsymbol{n} = qv_{s,p}(p - p_0), \tag{2.12}$$

where $\boldsymbol{n}$ is the outward unit normal vector, and the surface recombination velocities are defined as

$$v_{s,n} = \frac{A_n^* T^2}{qN_c}, \tag{2.13}$$

$$v_{s,p} = \frac{A_p^* T^2}{q N_v}. \tag{2.14}$$

The equilibrium carrier concentrations at the interface are expressed as

$$n_0 = N_c \exp\left(-\frac{q\phi_B}{k_B T}\right), \tag{2.15}$$

$$p_0 = N_v \exp\left(-\frac{q(E_g - \phi_B)}{k_B T}\right), \tag{2.16}$$

where N_c, N_v are the effective densities of states in the conduction and valence bands, respectively, and E_g is the bandgap.

These expressions couple the carrier concentrations at the semiconductor surface with the Schottky barrier, thereby enforcing the rectifying behavior at the interface within the DDM.

2.2.2 Ohmic Contact Theory

In contrast to Schottky contacts, an Ohmic contact allows majority carriers to flow freely between the metal and the semiconductor with negligible barrier resistance. As shown in Figure 3 b) for n-type and d) for p-type semiconductors, respectively, when the thermodynamic equilibrium is established, the Fermi level of the metal aligns with that of the semiconductor without significant band bending.

In numerical simulations, Ohmic contacts are typically treated as Dirichlet boundary conditions for both potential and carrier densities. The potential at the contact is just the bias applied externally plus the built-in potential:

$$\mathrm{V} = \mathrm{V}_{\mathrm{eq}} + U_{bias} + V_{built-in}, \tag{2.17}$$

where V_{eq} is the equilibrium reference level of the potential indication, U_{bias} is the external bias applied on the electrode and $V_{built-in}$ the built-in potential.

And the carrier concentrations are set to their thermal equilibrium values:

$$n = n_0 = \frac{N_c \exp(E_F - E_c)}{k_B T}, \tag{2.18}$$

$$p = p_0 = \frac{N_v \exp(E_v - E_F)}{k_B T}, \tag{2.19}$$

with E_F representing the equilibrium Fermi level. These conditions ensure that Ohmic contacts act as ideal reservoirs for electrons and holes, providing efficient injection and extraction of carriers into the device without introducing artificial barriers.

2.3 Drift-Diffusion Model

With the previous two sections' introduction, the DDM to be studied can be elaborated here. It describes carrier transport by combing electrostatics with continuity relations for electrons and holes, accounting for both drift in the electric field and diffusion due to concentration gradients [12]. In the context of planar GaAs Schottky diodes, the DDM provides an adequate balance between physical accuracy and computational efficiency.

The governing equations are then the Poisson equation eq. (2.1), describing the electrostatic potential, the drift-diffusion relations equations (2.2) and (2.3), describing the carrier current densities and the continuity equations (2.6) and (2.7), enforcing charge conservations. They form a coupled system of

nonlinear partial differential equations (PDEs) [12]. Analytical solutions are thus not feasible except for highly idealized cases. Hence a numerical approach based on FEM is employed in this work and will be introduced in the next section.

And to obtain a well-posed system, appropriate boundary conditions must be specified:

- At Schottky contacts, carrier densities and currents are constrained by the barrier height and thermionic emission processes, see equations (2.10-12) .
- At Ohmic contacts, the potential is fixed by the externally applied bias and carrier concentrations are set to their equilibrium values, see equations (2.17-19).
- Other boundaries than the two contacts are viewed as insulating boundaries. Neumann conditions are imposed to ensure zero normal current flow.

The DDM defined above provides the complete physical description of the planar GaAs Schottky diode. It assumes drift-diffusion dominated transport under local quasi-equilibrium conditions, while neglecting quantum effects such as tunneling or ballistic transport.

2.4 Finite Element Method (FEM) Formulation

2.4.1 Normalization

Before deriving the weak form of DDM, it is advantageous to normalize the same equations [13]. This procedure expresses all variables in dimensionless form, reduces the disparity in their numerical ranges, and improves the conditioning of the finite element matrices. Without normalization, the coexistence of quantities such as carrier densities on the order of $10^{24}\ cm^{-3}$ and electrostatic potentials of only a few tenths of a volt would result in poorly scaled matrices and hinder numerical convergence.

The normalization factors introduced in Table 2.1 are to define dimensionless variables. After solving the normalized system, the physical quantities of interest can be recovered by multiplying with the appropriate factors.

Table 2.1 Normalization coefficients.

Quantity ($V_{real} = V_{norm} \cdot Factor$)	Normalized quantity symbol	Normalization factor
Time	t	$\frac{L_0^2}{D_0}$
Position	r	$L_0 = \sqrt{\frac{\varepsilon V_t}{q n_i}}$
Electric scalar potential	ϕ	$V_t = \frac{k_B T}{q}$
Electric field	$E = -\nabla\phi$	$\frac{V_t}{L_0}$
Electron density	n	n_i
Hole density	p	n_i
Carrier diffusion constants	$\gamma_n^{-1}, \gamma_p^{-1}$	$D_0 = D_p$
Carrier mobilities	$\gamma_n^{-1}, \gamma_p^{-1}$	$\frac{D_0}{V_t}$
Current densities	j_n, j_p	$J_0 = \frac{q n_i D_0}{L_0}$

In the definitions, n_i is the intrinsic carrier concentration and the hole diffusion constant is chosen to be the reference, without loss of generality.

The Poisson equation can be normalized in the following way:

$$\frac{\varepsilon V_t}{q n_i}\left(\frac{\Delta\varphi(\boldsymbol{r},t)}{V_t}\right) = \frac{n(\boldsymbol{r},t)}{n_i} - \frac{p(\boldsymbol{r},t)}{n_i} + \frac{N_A(\boldsymbol{r})}{n_i} - \frac{N_D(\boldsymbol{r})}{n_i}. \tag{2.20}$$

Then, the following dimensionless normalized quantities can be introduced:

$$\phi(\boldsymbol{r},t) = \frac{\varphi(\boldsymbol{r},t)}{V_t},$$

$$n_n(\boldsymbol{r},t) = \frac{n(\boldsymbol{r},t)}{n_i}, p_n(\boldsymbol{r},t) = \frac{p(\boldsymbol{r},t)}{n_i}, N_A^n(\boldsymbol{r}) = \frac{N_A(\boldsymbol{r})}{n_i}, N_D^n(\boldsymbol{r}) = \frac{N_D(\boldsymbol{r})}{n_i},$$

$$\boldsymbol{r}_n = \frac{\boldsymbol{r}}{\sqrt{\frac{\varepsilon V_t}{q n_i}}} = \frac{\boldsymbol{r}}{L_0}, t_n = \frac{t}{\tau},$$

deriving the normalized Poisson equation:

$$\Delta\phi(\boldsymbol{r}_n, t_n) = n_n(\boldsymbol{r},t) - p_n(\boldsymbol{r},t) + N_A^n(\boldsymbol{r}) - N_D^n(\boldsymbol{r}) = -\rho_n, \tag{2.21}$$

where ρ_n is the normalized net charge concentration.

The current flow equation can also be normalized, without loss of generality, the electrons current is shown here:

$$\frac{1}{\frac{qn_i}{L_0}D_0}\boldsymbol{J}_n(\boldsymbol{r},t) = -\frac{V_t}{D_0}\frac{\mu_n n(\boldsymbol{r},t)}{n_i}L_0\nabla\frac{\varphi(\boldsymbol{r},t)}{V_t} + \frac{D_n(\boldsymbol{r})}{D_0}L_0\frac{\nabla n(\boldsymbol{r},t)}{n_i}. \tag{2.22}$$

With $j_n = \frac{J_n}{J_0}$, the normalized form becomes:

$$\boldsymbol{j}_n(\boldsymbol{r}_n,t_n) = \gamma_n^{-1}(\boldsymbol{r}_n)[-n_n(\boldsymbol{r}_n,t_n)\nabla_n\phi(\boldsymbol{r}_n,t_n) + \nabla_n n_n(\boldsymbol{r}_n,t_n)]. \tag{2.23}$$

Similarly, the hole current equation can also be normalized:

$$\boldsymbol{j}_p(\boldsymbol{r}_n,t_n) = \gamma_p^{-1}(\boldsymbol{r}_n)[-p_n(\boldsymbol{r}_n,t_n)\nabla_n\phi(\boldsymbol{r}_n,t_n) - \nabla_n p_n(\boldsymbol{r}_n,t_n)]. \tag{2.24}$$

Following this manner, the continuity equations can be normalized as

$$\frac{\partial}{\partial t_n}n_n(\boldsymbol{r}_n,t_n) = \nabla_n\boldsymbol{j}_n(\boldsymbol{r}_n,t_n), \tag{2.25}$$

$$\frac{\partial}{\partial t_n}p_n(\boldsymbol{r}_n,t_n) = -\nabla_n\boldsymbol{j}_p(\boldsymbol{r}_n,t_n). \tag{2.26}$$

For simplicity, in this study, the generation and recombination are neglected.

The normalization procedure transforms the original drift-diffusion equations into a dimensionless system. This significantly improves the numerical stability and efficiency of the finite element implementation.

2.4.2 Weak Formulation

The DDM, defined in Section 2.3, consists of a system of coupled nonlinear PDEs that cannot be solved analytically for realistic device geometries. To enable numerical simulation, the equations are reformulated in their weak form, which provides the foundation of the FEM. The weak formulation is obtained by multiplying the governing equations by test functions and integrating over the computational domain.

The Poisson equation reads as eq. (2.1). Multiplying both sides by a test function w and integrating over the domain Ω:

$$\int_\Omega w\,\nabla\cdot\nabla\phi\,d\Omega = -\int_\Omega w\,\rho_n\,d\Omega\,. \tag{2.27}$$

Applying the divergence theorem and separating the boundary integral:

$$-\int_\Omega \nabla w\cdot\nabla\phi\,d\Omega + \int_{\partial\Omega} w\nabla\phi\cdot\boldsymbol{n}dS = -\int_\Omega w\,\rho_n\,d\Omega\,, \tag{2.28}$$

where $\boldsymbol{n}$ denotes the outward normal on the boundary $\partial\Omega$.

Depending on the type of boundary, at Schottky or Ohmic contacts, there are Dirichlet boundary conditions, the potential ϕ is prescribed and the boundary integral is eliminated. For the other insulating surfaces, the Neumann boundary condition is applied, $\nabla\phi\cdot\boldsymbol{n} = 0$ ensures the surface integral vanishes. Thus, the weak form of the Poisson equation is:

$$\int_\Omega \nabla w\cdot\nabla\phi\,d\Omega = \int_\Omega w\rho_n d\Omega. \tag{2.29}$$

For electrons, the continuity equations are equations (2.6) and (2.7), complemented with the current density equations (2.2) and (2.3). The weak form is obtained by multiplying by a test function w_n and integrating:

$$\int_{\Omega} w_{\mathrm{n}} \frac{\partial n_n}{\partial t_n} \, d\Omega \; = \int_{\Omega} w_n \nabla \cdot \boldsymbol{j}_{\mathrm{n}} \, d\Omega. \tag{2.30}$$

Applying integration by parts to the current term:

$$\int_{\Omega} w_{\mathrm{n}} \frac{\partial n_n}{\partial t_n} \, d\Omega \; = -\int_{\Omega} \nabla w_{\mathrm{n}} \cdot \boldsymbol{j}_{\mathrm{n}} \, d\Omega \; + \int_{\partial\Omega} w_{\mathrm{n}} \, (\boldsymbol{j}_{\mathrm{n}} \cdot \boldsymbol{n}) dS. \tag{2.31}$$

Boundary terms are determined by the contact conditions. At Schottky contacts, $j_n \cdot \boldsymbol{n}$ is defined by the thermionic emission boundary condition. At Ohmic contacts, n is fixed to its equilibrium value . And at insulating boundaries, $\boldsymbol{j}_n \cdot \boldsymbol{n} = 0$.

The weak form of the hole continuity equation is derived analogously with test function w_p:

$$\int_{\Omega} w_{\mathrm{p}} \frac{\partial p_n}{\partial t_n} \, d\Omega \; = -\int_{\Omega} \nabla w_{\mathrm{p}} \cdot \boldsymbol{j}_{\mathrm{p}} \, d\Omega \; + \int_{\partial\Omega} w_{\mathrm{p}} \, (\boldsymbol{j}_{\mathrm{p}} \cdot \boldsymbol{n}) dS. \tag{2.32}$$

Boundary conditions determine the contribution of the surface integrals to the element equations. The Schottky contacts introduce Robin-type boundary conditions for electron and hole fluxes:

$$\boldsymbol{j}_n \cdot \boldsymbol{n} = -q v_{s,n} (n - n_0), \tag{2.33}$$

$$\boldsymbol{j}_p \cdot \boldsymbol{n} = q v_{s,p} (p - p_0), \tag{2.34}$$

which directly modify the boundary flux vector as well as the stiffness matrix.

The Ohmic contacts impose Dirichlet conditions for potential and carrier densities: $\phi = V_0, n = n_0, p = p_0$, which are enforced by fixing nodal values and adjusting the corresponding rows of the global system matrix.

The insulating boundaries apply Neumann conditions with zero flux, that is, $\nabla\phi \cdot \boldsymbol{n} = 0, \boldsymbol{j}_n \cdot \boldsymbol{n} = 0, \boldsymbol{j}_p \cdot \boldsymbol{n} = 0$, thereby eliminating the boundary integrals in the element equations.

2.4.3 Discretization and Basis Functions

In FEM, the computational domain Ω is shown in Figure 4, which are just two rectangles representing the n- and n^+-type semiconductors, and the electrodes are modeled as some parts of the boundaries, the top electrode is marked in magenta and the side electrode in blue. Then Ω is discretized into a finite number of non-overlapping triangular subdomains called elements. To keep sufficient accuracy on some regions and save computational power on the others, an adapted meshing algorithm is developed. Finer meshes are assigned near the electrodes, as shown in Figure 5. The red region represents the n-region, while the blue one is for the n^+ region. The global solution of the drift-diffusion system is then approximated by a linear combination of locally defined basis functions.

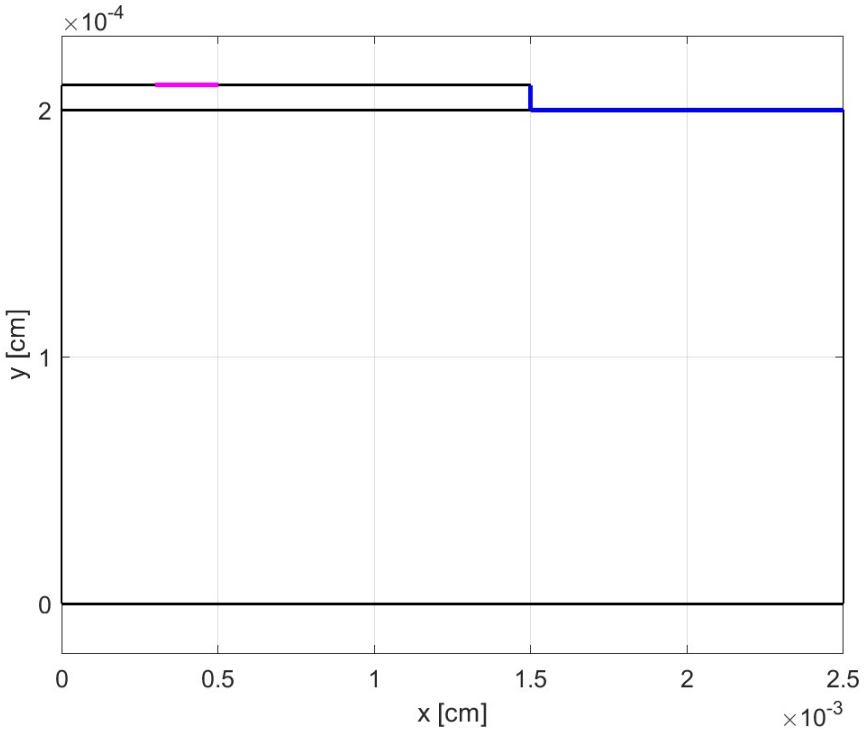


Figure 4. Computational domain, with magenta marking the top electrode and blue marking the side electrode.

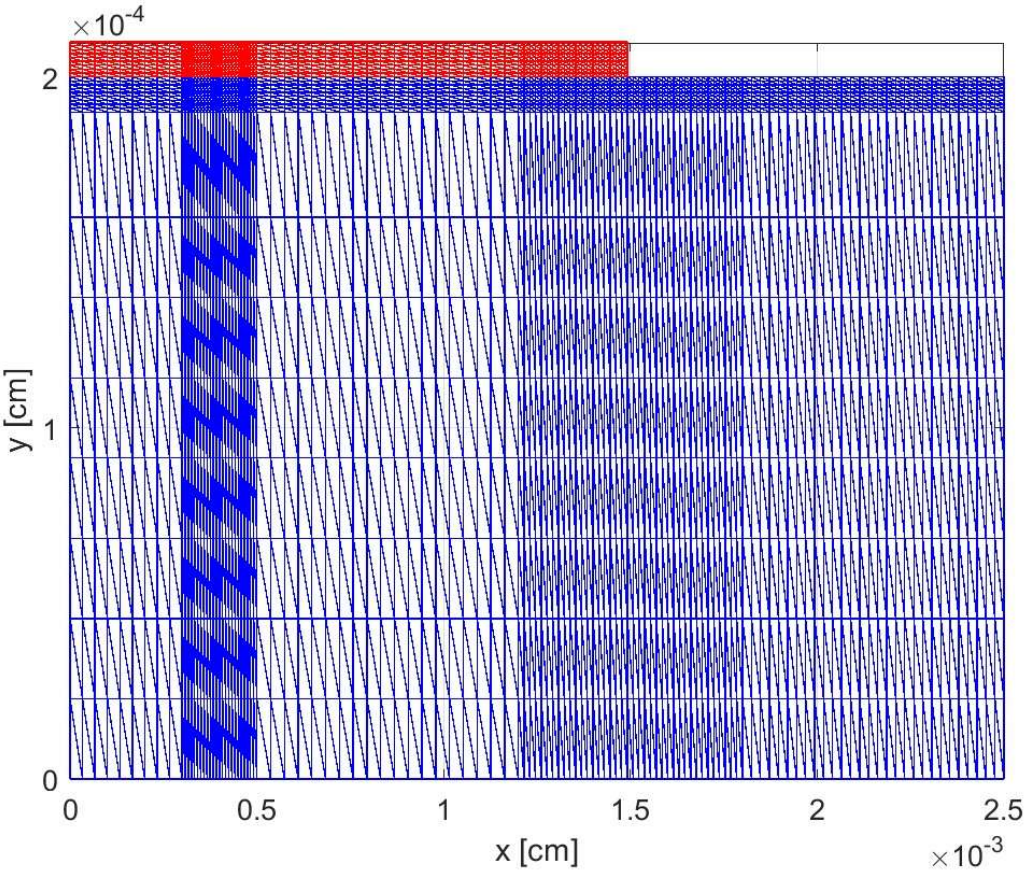


Figure 5. Mesh of the simulation domain, red represents the n-region and blue the n^+-region.

Let $\{N_i(x,y)\}$ denote the set basis functions associated with the nodes of the mesh. For a linear triangular element, the N_i are piecewise linear functions that take the value 1 at the node i, and 0 at the other nodes of the element. The electrostatic potential $\varphi(x,y)$ is then approximated as

$$\phi(x,y) \approx \sum_{j=1}^{n_{nodes}} \phi_j N_j(x,y), \tag{2.35}$$

where ϕ_j are the nodal values of the potential. Similarly, the carrier concentrations are approximated by

$$n_n(x,y) \approx \sum_{j=1}^{n_{nodes}} n_j N_j(x,y), \tag{2.36}$$

$$p_n(x,y) \approx \sum_{j=1}^{n_{nodes}} p_j N_j(x,y)\,. \tag{2.37}$$

Substituting these interpolations into the weak form of the Poisson and continuity equations leads to systems of algebraic equations, in which the unknowns are the nodal values φ_j, n_j, p_j. The DDM is solved using the Galerkin method, where the set of test functions w, w_n, w_p is identical to the set of shape functions. This ensures that the resulting system matrix is symmetric and positive definite. The Kronecker delta property of the basis function $N_i(x_j) = \delta_{ij}$ guarantees that the unknown coefficients directly represent nodal values, thereby simplifying interpolation and assembly. The element-level contribution to the system matrix is computed by integrating products of shape functions and their gradients over each element. For example, the element stiffness matrix associated with the Poisson equation is given by

$$K_{ij}^e = \int_{\Omega_e} \nabla N_i \cdot \nabla N_j d\Omega, \tag{2.38}$$

where Ω_e denotes the domain of element e.

2.4.4 Element Matrices

As shown in the previous section, the weak form of the Poisson and continuity equations can be expressed in terms of element-level matrices and vectors. These element matrices represent the local contributions of each element to the global system.

For the Poisson equation, the weak form is

$$\int_{\Omega^e} \nabla N_i \cdot \nabla \phi \, d\Omega = -\int_{\Omega^e} N_i \rho_n \, d\Omega\,. \tag{2.39}$$

Expanding $\phi^e(x,y) \approx \sum_{j=1}^{3} \phi_j^e N_j^e(x,y)$, the element stiffness matrix is obtained:

$$K_{ij}^e = \int_{\Omega^e} \nabla N_i \cdot \nabla N_j \, d\Omega\,, \tag{2.40}$$

and the element load vector is

$$f_i^e = -\int_{\Omega^e} N_i \rho_n \, d\Omega. \tag{2.41}$$

Thus, the discrete element equation reads

$$\sum_j K_{ij}^e \, \phi_j \; = \; f_i^e. \tag{2.42}$$

For electrons, the weak form yields

$$\int_{\Omega^e} N_i \, \frac{\partial n_n}{\partial t_n} \, d\Omega \; + \; \int_{\Omega^e} \nabla N_i \; \cdot \; \boldsymbol{j}_\mathrm{n} \, d\Omega \; = \; \int_{\partial\Omega^e} N_i \, (\boldsymbol{j}_\mathrm{n} \; \cdot \; \boldsymbol{n}) \, dS. \tag{2.43}$$

Substituting the interpolations $n_n^e(x,y) \approx \sum_{j=1}^3 n_j^e N_j(x,y)$, and the expression for the normalized electron current $\boldsymbol{j}_n = \gamma_n^{-1}(-n\nabla\phi + \nabla n)$, the discretized terms can be written as:

the Mass matrix:

$$M_{ij}^e \; = \; \int_{\Omega^e} N_i \, N_j \, d\Omega \, , \tag{2.44}$$

the drift-diffusion matrix:

$$C_{ij}^e = \int_{\Omega^e} \nabla N_i \; \cdot \left(-\gamma_n^{-1} n N_j \nabla\phi + \gamma_n^{-1} \nabla N_j\right) d\Omega, \tag{2.45}$$

and the boundary flux vector:

$$b_{n,i}^e \; = \; \int_{\partial\Omega^e} N_i \, (\boldsymbol{j}_\mathrm{n} \; \cdot \; \boldsymbol{n}) \, dS. \tag{2.46}$$

The electron continuity equation at element level is then expressed as

$$M_n^\mathrm{e} \left(\frac{dn^\mathrm{e}}{dt}\right) + \; C^e \, (\phi^\mathrm{e}, n^\mathrm{e}) = \; b_n^\mathrm{e}. \tag{2.47}$$

Similar expressions hold for the hole continuity equation

$$M_p^\mathrm{e} \left(\frac{dp^\mathrm{e}}{dt}\right) + \; C^\mathrm{e} \, (\phi^\mathrm{e}, p^\mathrm{e}) = \; b_p^\mathrm{e}. \tag{2.48}$$

2.4.5 Global Assembly

The element matrices and vectors derived in the previous section represent the local contributions of individual finite elements. To obtain the complete system of equations describing the device, these local contributions must be assembled into global matrices according to the finite element mesh connectivity.

For example, the local stiffness matrix of the Poisson equation,

$$K_{ij}^e \; = \; \int_{\Omega^e} \nabla N_i \; \cdot \; \nabla N_j \, d\Omega \, , \tag{2.49}$$

is inserted into the corresponding position of the global stiffness matrix K. The same procedure applies to the mass matrices, drift-diffusion matrices and load vectors obtained from the continuity equations. After assembly, the global system of equations can be written in compact forms as

$$K\phi = F, \tag{2.50}$$

$$M_n \frac{dn}{dt} + C(\phi, n) + Kn = \mathrm{b_n}, \tag{2.51}$$

$$M_p \frac{dp}{dt} + C(\phi, p) + Kp = \mathrm{b_p}. \tag{2.52}$$

The matrices K, C, M_n, M_p and the vectors F, b_n, b_p are assembled from the corresponding element-wise contributions. For example, the global stiffness matrix K consists of entries K_{ij}, similarly for the others.

2.4.6 Time Discretization

In the transient case, the continuity equations for electrons and holes contain time derivative terms, which must be discretized before numerical solution. Among various schemes, the backward Euler method [14] is adopted in this work, due to its unconditional stability and robustness for stiff problems. For electrons, the weak form leads to the semi-discrete equation

$$M_n \frac{dn}{dt} + C(\phi, n) = b_n, \tag{2.53}$$

where M_n is the mass matrix, C represents the drift-diffusion contributions, and b_n contains the boundary fluxes and source terms. Applying the backward Euler method with time step Δt, the fully discrete form becomes

$$M_n \frac{n^{k+1} - n^k}{\Delta t} + C\left(\phi^{k+1}, n^{k+1}\right) = b_n^{k+1}, \tag{2.54}$$

where n^k and n^{k+1} denote the nodal electron concentrations at the previous and current time steps, respectively. An analogous expression holds for holes.

2.4.7 Nonlinear Iterative Solver

After spatial and temporal discretizations, the drift-diffusion system reduces to a set of nonlinear algebraic equations that couple the electrostatic potential, electron and hole densities. This work adopts a Gummel-type decoupled iteration combined with a successive under-relaxation method. This approach offers improved robustness and significantly reduced computational complexity, at the expense of slower convergence.

The Gummel method decouples the nonlinear system into subproblems that are solved sequentially within each iteration cycle:

1. Electrostatic update:

   Solve the Poisson equation for the potential ϕ^{i+1} using the current estimates of the carrier densities.

2. Carrier update:

   With the updated potential, solve the electron and hole continuity equations for n^{i+1} and p^{i+1}.

3. Matrix update:

   Reassemble the drift-diffusion matrices using the new potential and carrier distributions.

4. Relaxed update:

   Apply under-relaxation to stabilize convergence:

   $$u^{i+1} = u^i + \alpha\left(\tilde{u}^{i+1} - u^i\right),$$

   where $\tilde{u}^{i+1}$ is the newly computed value, u^i is the previous iterate, and $\alpha \in (0,1]$ is the relaxation parameter.

5. Convergence check:

The iteration continues until the residual norm satisfies

$$\max\left[\frac{\left(\tilde{u}^{i+1}-u^{i}\right)}{u^{i}}\right]<tol,$$

with tol a prescribed tolerance.

2.5 Summary of Theoretical Framework

In this chapter, the theoretical foundation for the numerical modeling of planar GaAs Schottky diodes has been established. First, the DDM was introduced as the governing framework, together with the boundary conditions arising from the Schottky and Ohmic contacts. The equations were then normalized to obtain a dimensionless form, ensuring improved numerical stability.

Subsequently, the FEM was applied to derive the weak formulations of the Poisson and continuity equations. Using appropriate basis functions, the weak forms were discretized into element-level matrices, which were assembled into the global system of equations.

Finally, the treatment of the transient problem was discussed using the backward Euler time discretization, and the nonlinear system was solved by means of a Gummel-type iteration combined with the successive under-relaxation method.

Together, these steps provide a consistent and robust theoretical framework for the numerical implementation.

3 Results

This chapter presents the numerical results obtained from the finite element implementation of the DDM. While the theoretical framework and numerical formulation were established in the last chapter, the purpose here is to demonstrate the performance of the solver on representative device configurations and to validate the implemented model. And the descriptions and discussions will be mainly focused on the comparison between the results produced by the in-house developed code and the reference from COMSOL.

3.1 Simulation Setup

The computational domain and mesh discretization have already been introduced in Section 2.4.3 (see Figure 4 and 5). The same domain and meshing strategy are employed in the simulations presented here. Two types of contact configurations are considered:

1. Ohmic-Ohmic case, in which both electrodes are treated as ideal Ohmic contacts.
2. Schottky-Ohmic case, where the electrode at top of the n-region is treated as a Schottky contact, while the other one as an Ohmic contact.

The external bias was applied only to the electrode at the top and varied over a range of 0 V to 1 V. The current-voltage ($I-V$) characteristics were recorded. The simulations employed a uniform time step of $\Delta t \approx 6.8 * 10^{-16} s$, a relaxation factor of $\alpha = 0.5$, and a convergence tolerance of $tol = 10^{-5}$.

3.2 Ohmic-Ohmic Contact Results

As a first benchmark, the diode was simulated with Ohmic contacts applied to both electrodes.

3.2.1 $I-V$ characteristics

This configuration is expected to yield a nearly linear $I-V$ characteristic, since no potential barrier is presented at the contact. Given the drift and diffusion current density equations (2.2) and (2.3), the value in each element is evaluated, and the elements comprising the electrode are used for the current calculation. To compute the total current I across the electrode, the local current density vector $J = (J_x, J_y)$ onto the local outward unit normal vector $n = (n_x, n_y)$ of each electrode unit, and sum over the electrode length:

$$I = \sum_{electrode} J \cdot n \cdot L = \sum_{electrode} (J_x n_x + J_y n_y) L, \tag{3.1}$$

with L the electrode length. This expression approximates the normal component of the current across each electrode element and sums the contributions to obtain the total current across the electrode.

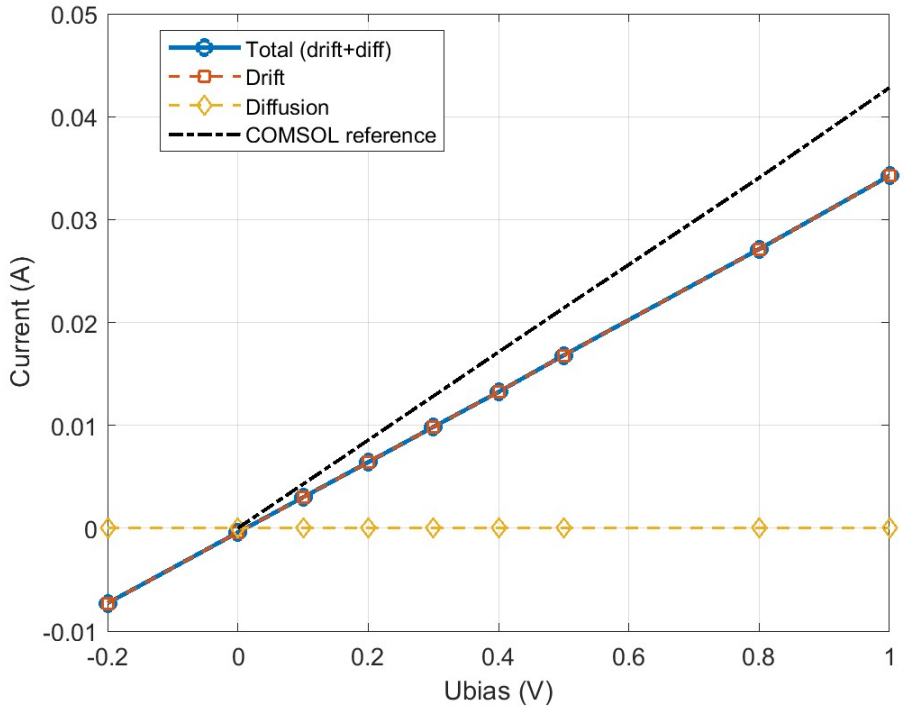


Figure 6. Current under different biases evaluated at the top Ohmic contact.

As shown in Figure 6, the calculated current curve is straight as expected, while the corresponding values for higher bias voltages deviate from the COMSOL reference. On the one hand, the accuracy of the calculation is not high enough. For instance, in the drift current (2.2), the average carrier densities over the element are simply applied instead of a more carefully handled Gauss integration. On the other hand, the COMSOL reference current may also be affected by meshing. As shown in Figure 7, the computed electric field along the top contact shows a strong dependence on the mesh resolution (i.e. 'Fine' vs. 'Extra Fine'), which may indicate that the reference current is also sensitive to meshing and may be numerically inaccurate.

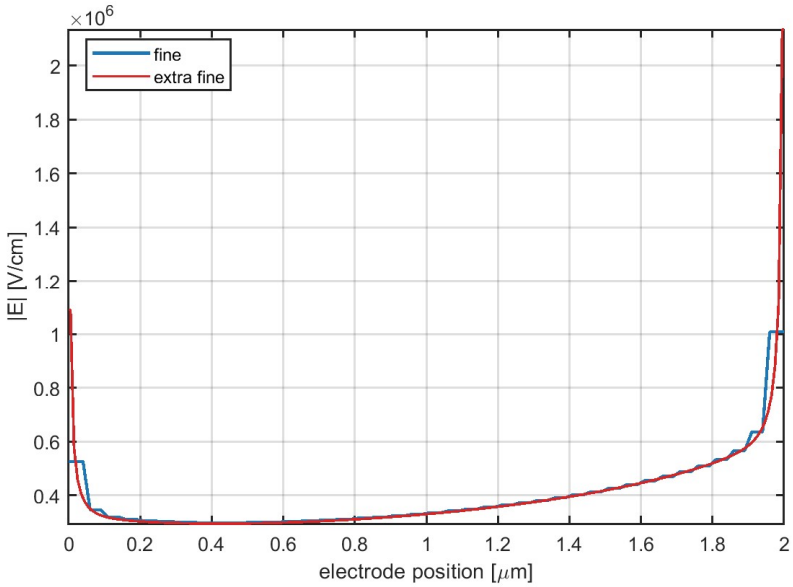


Figure 7. Electric field profiles with different meshing methods.

3.2.2 Potential distribution

The potential distributions of the structure under study under 0 V, 0.5 V, 1 V applied voltages are shown in Figure 8. If no external voltage source is applied, the potential remains constant in each region of semiconductor. When a forward bias of 0.5 V is applied, a nearly linear potential drop is observed across the semiconductor bulk, as a result of the solution to the Poisson equation. With a higher bias, the potential drop becomes more pronounced, while the distribution is still linear.

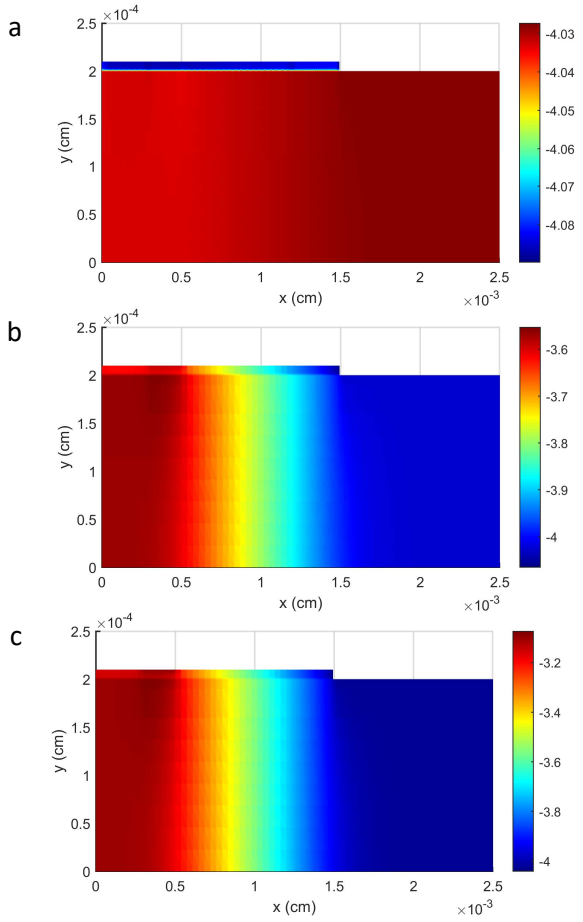


Figure 8. Potential distribution under a) 0V, b) 0.5V and c) 1V bias.

3.2.3 Electric field distribution

The electric field distributions under different biases are also shown for the studied structure. As expected, the corresponding electric field is nearly uniform throughout the whole diode, under Ohmic-Ohmic contacts, with only slight variation around the electrode boundaries, especially the one with the applied bias. At 0 bias, the field is close to zero within the device, it only exists at the semiconductor interface, due to the potential difference generated by different doping concentrations. Increasing the bias primarily changes the magnitude of the field rather than its spatial distribution, as observed in Figure 9 b) and c).

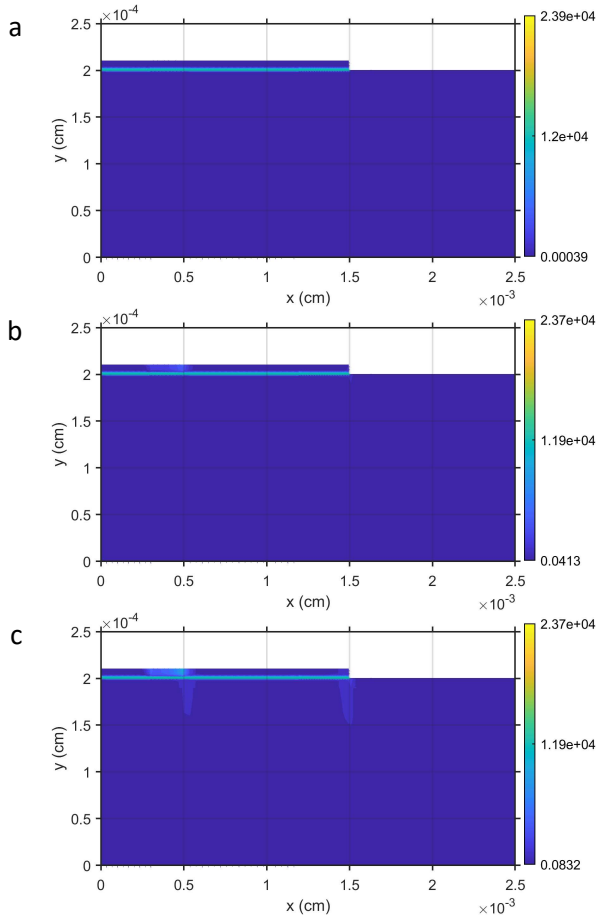
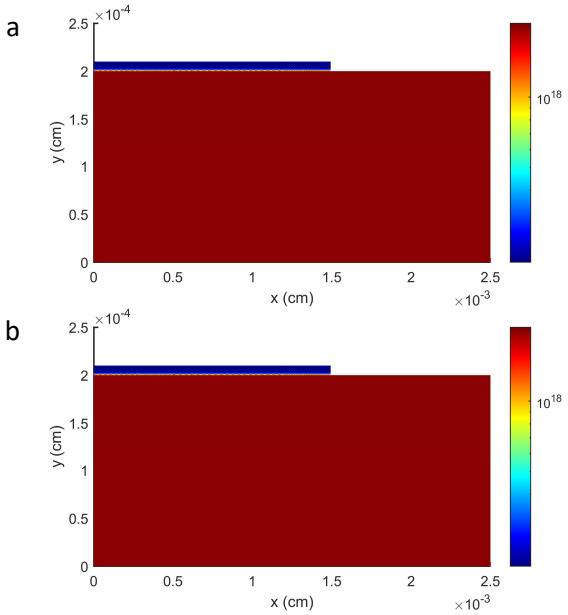


Figure 9. Norm of the electric field distribution under a) 0V, b) 0.5V and c) 1V bias.

3.2.4 Charge distribution

In the n/n^+ structure, since the hole concentration is negligible, only the electrons concentration is presented in this section. The carrier concentration remains nearly uniform across the device. The associated results agree well with the COMSOL reference, with only minor discrepancies around the n/n^+ interface, due to the mesh resolution.



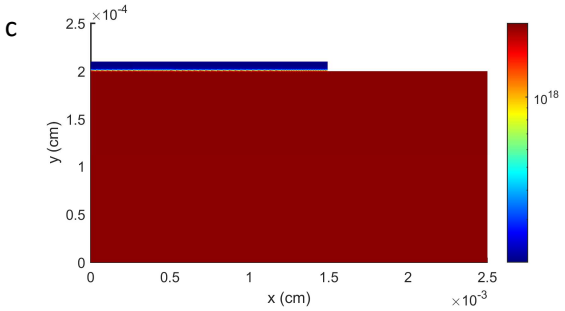


Figure 10. Electrons distribution under a) 0V, b) 0.3V and c) 1V bias.

3.2.5 Cutline profiles

To better quantitatively compare the simulation results to the COMSOL reference, the electron concentration and potential distributions under different biases are extracted along the cutlines, specified by the red dashed lines in Figure 11.

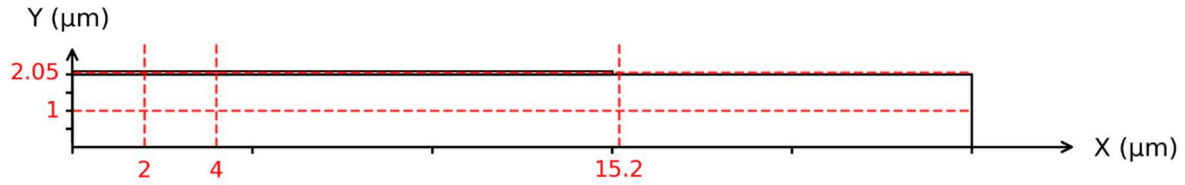


Figure 11. Cutline positions.

Due to space limitations, only a subset of the cutline comparison results are shown in the main text. The complete set of cutlines can be found in Appendix B:.

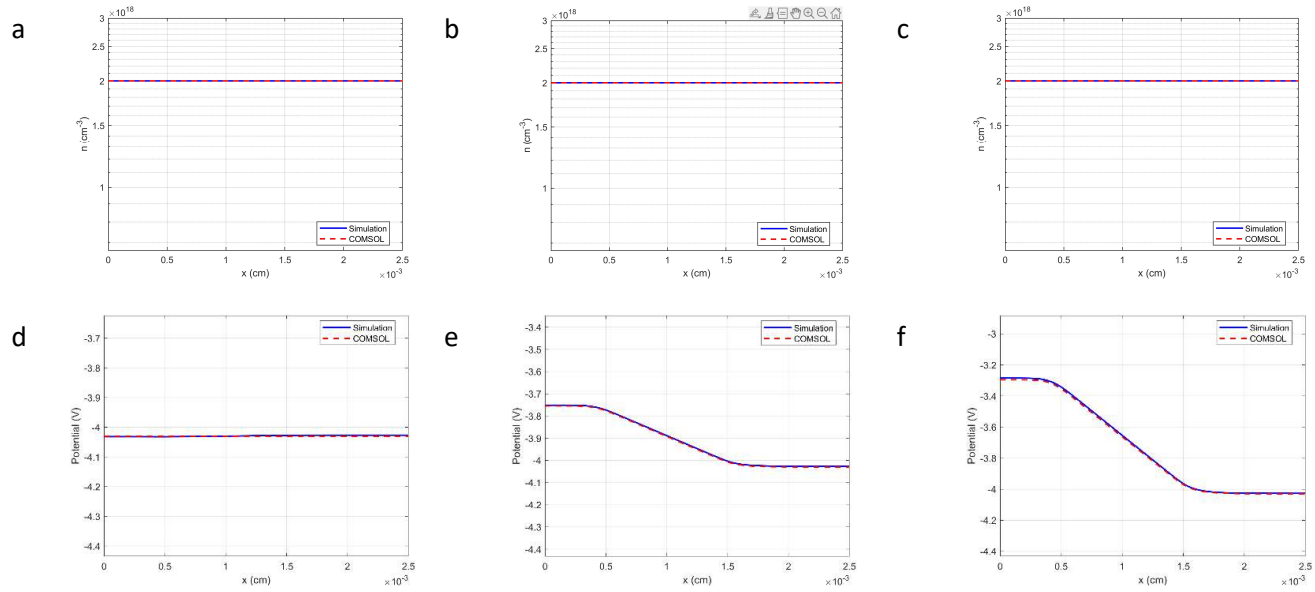


Figure 12. Cutline profiles of electron concentration and potential at $y = 1$ μm under Ohmic contact conditions. a) – c) are electron concentration distributions and d) – f) are potential distributions under 0V, 0.3V, 0.8V, respectively.

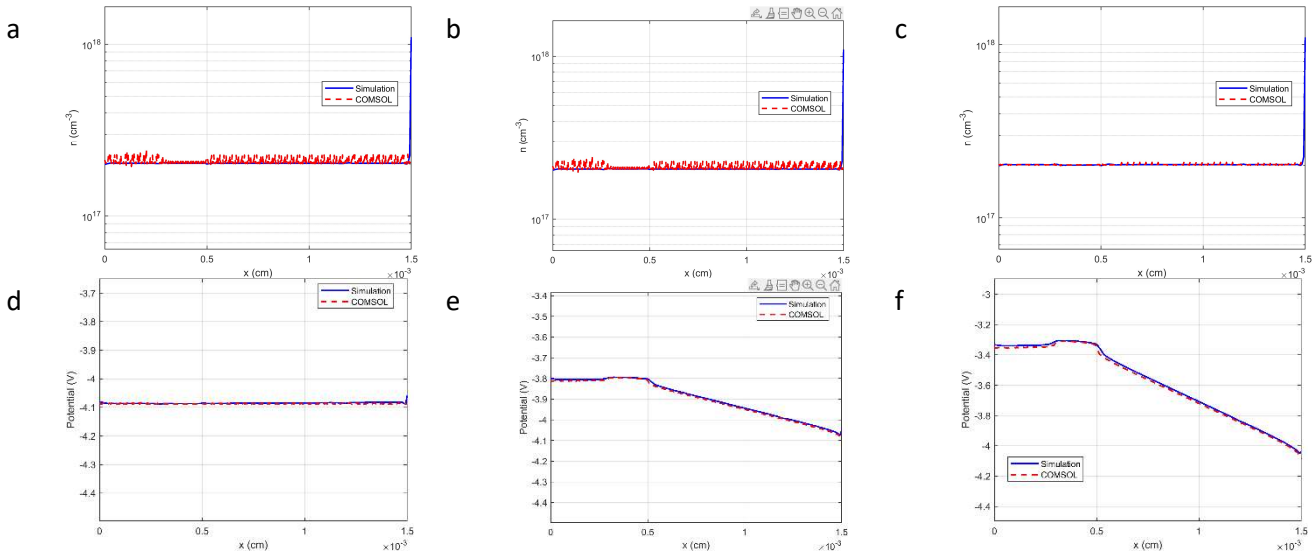


Figure 13. Cutline profiles of electron concentration and potential at $y = 2.05$ μm under Ohmic contact conditions. a) – c) are electron concentration distributions and d) – f) are potential distributions under 0V, 0.3V, 0.8V, respectively.

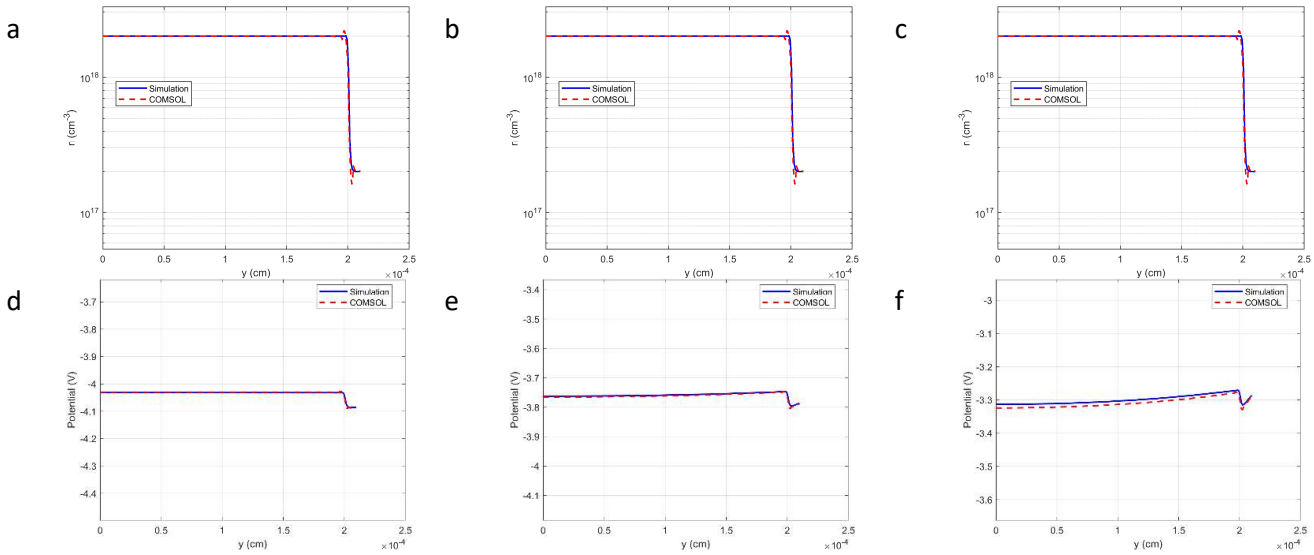


Figure 14. Cutline profiles of electron concentration and potential at $x = 2$ μm under Ohmic contact conditions. a) – c) are electron concentration distributions and d) – f) are potential distributions under 0V, 0.3V, 0.8V, respectively.

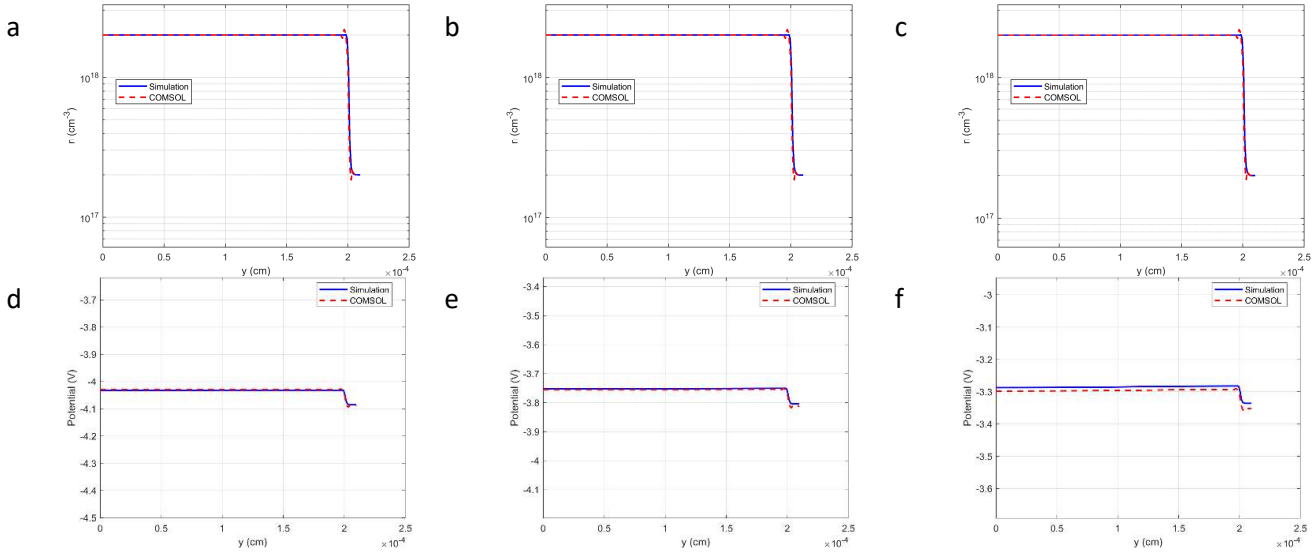


Figure 15. Cutline profiles of electron concentration and potential at $x = 4\ \mu m$ under Ohmic contact conditions. a) – c) are electron concentration distributions and d) – f) are potential distributions under 0V, 0.3V, 0.8V, respectively.

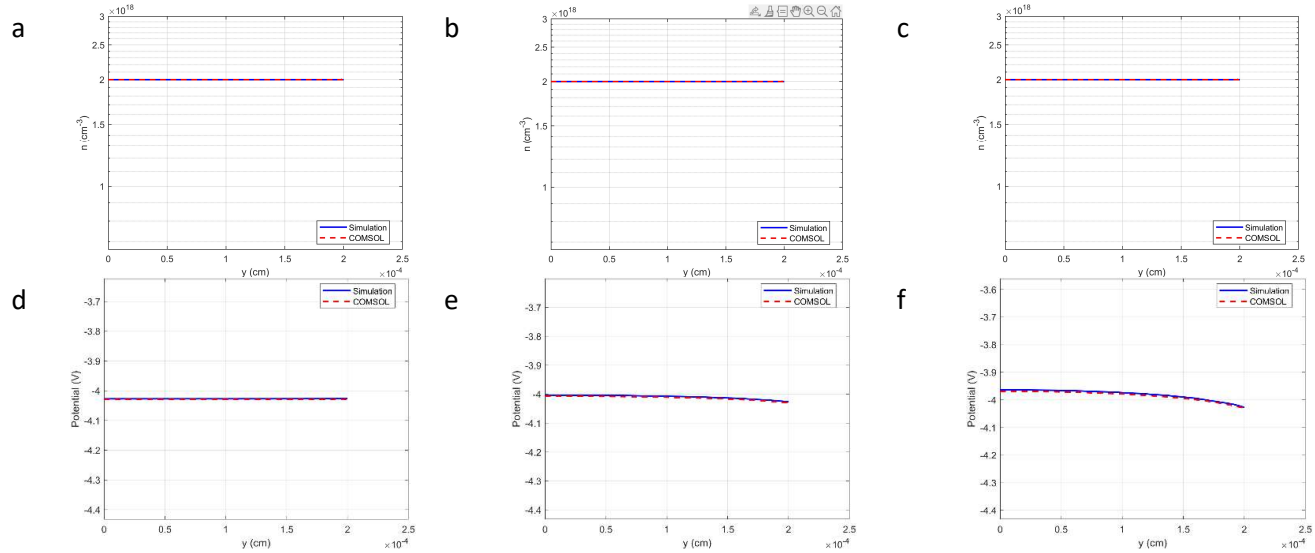


Figure 16. Cutline profiles of electron concentration and potential at $x = 15.2\ \mu m$ under Ohmic contact conditions. a) – c) are electron concentration distributions and d) – f) are potential distributions under 0V, 0.3V, 0.8V, respectively.

In order to ensure a consistent comparison with COMSOL, the applied Dirichlet boundary conditions of the potential in the simulations were adjusted to match the ones of COMSOL, including the built-in potentials This numerical adjustment affects only the absolute reference of the electric potential, not the other physical quantities of interest. The reason is that the carrier concentrations, electric field and current densities depend on the potential difference and gradients rather than their absolute values.

Figure 14, 15 and 16 show the vertical cutline profiles at the left, middle, and right positions of the electrode, under applied biases of 0 V, 0.3 V, and 0.8 V. In all cases, the electron concentration remains essentially constant across the device thickness, and the simulated results overlap almost perfectly with the COMSOL curves. The potential profiles display nearly linear drops across the domain, with the slope increasing proportionally to the applied bias. No depletion regions are observed, which is consistent with the expected behavior of the device with the Ohmic contacts at both terminals.

Horizontal cutline profiles taken within the n and n^+ regions are presented in Figure 12 and 13 in the n-region, where the bias voltage is directly applied at the device surface, the potential exhibits a nearly linear gradient across the lateral direction. As the applied bias increases, the slope of the potential profile becomes steeper, reflecting the stronger electric field across the device. The electron concentration remains essentially uniform, as expected for Ohmic contacts without barrier formation. The comparison with COMSOL results shows a very good agreement, with only minor deviations at the electrode edges. In the n^+ region, the potential drops across the region. The slope increases as the applied voltage increases, which is consistent with the increased electric field in the bulk. The simulation results and COMSOL curves overlap almost perfectly, confirming the accuracy of the implemented solver.

The vertical and horizontal cutline analyses confirm that the Ohmic–Ohmic configuration produces the expected resistor-like behavior: constant carrier concentration, uniform electric field, and a linear potential drop across the device. The results match closely with COMSOL, validating the correctness of the implemented drift–diffusion solver under purely Ohmic boundary conditions.

3.3 Schottky-Ohmic Contact Results

The remaining results and plots are generated with the Schottky contact at the top, instead of the Ohmic one.

3.3.1 $I-V$ characteristics

Different from the last section, the currents are calculated with the thermionic emission boundary conditions (2.33) and (2.34), instead of the drift diffusion equations (2.2) and (2.3). This method accounts for the injection of carriers over the potential barrier at the contact and provides a more accurate description of the current through the device. As shown in Figure 17, the simulated $I-V$ curve is compared with the COMSOL reference. The simulation current remains close to zero under reverse and small forward biases, consistent with the blocking behavior of the Schottky contact. Once the forward bias exceeds about 0.3 V, the current rises rapidly, exhibiting the expected rectifying behavior. The simulation results agree well with the COMSOL reference, except for some minor deviations around the transition region. This agreement confirms the correct implementation of the thermionic emission boundary condition at the Schottky contact.

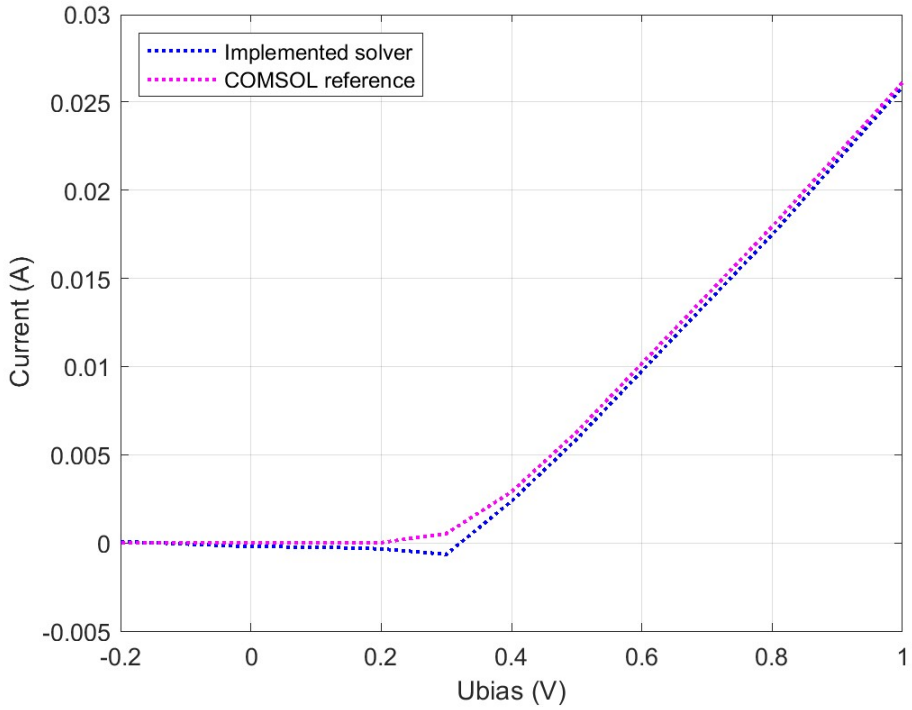


Figure 17. Current under different biases evaluated at the Schottky contact.

3.3.2 Potential distribution

The potential distributions of the structure under study are shown in Figure 18.

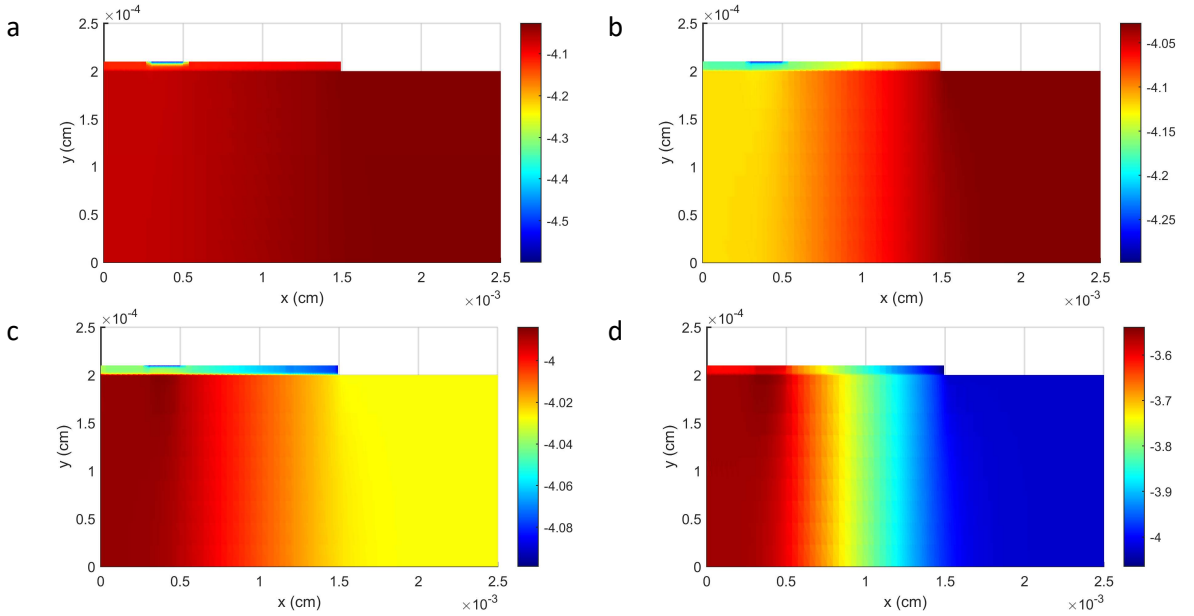


Figure 18. Potential distribution under a) 0V, b) 0.3V, c) 0.5V, and d) 1V.

When given no bias, a strong potential bending appears near the Schottky interface, indicating the formation of a depletion region and a built-in barrier. As the applied forward bias increases, this potential drop across the interface becomes smaller and gradually smooths out. At a high forward bias, the potential profile becomes nearly linear across the device, similar to the Ohmic case, because the Schottky barrier is almost completely flattened and the transport is dominated by the drift conduction.

3.3.3 Electric field distribution

Electric field distributions are shown in Figure 19 for the studied configuration.

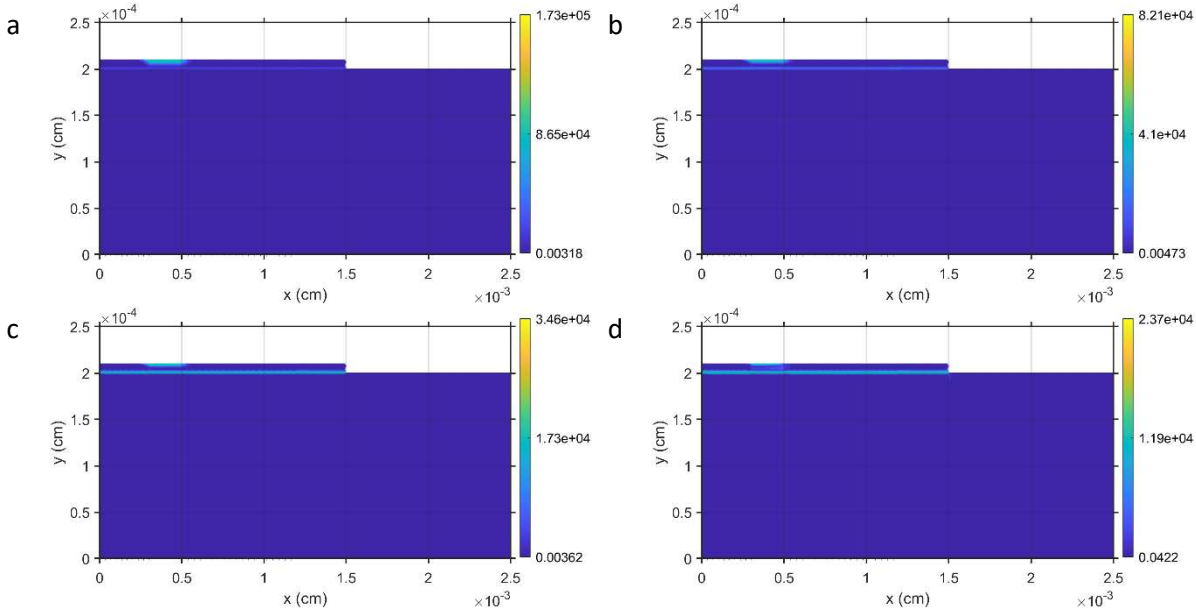


Figure 19. Electric field distribution under a) 0V, b) 0.3V, c) 0.5V and d) 1V bias.

At 0 V and 0.3 V, a strong localized electric field is observed near the Schottky interface, consistent with the depletion region formed at the equilibrium. The field magnitude reaches its maximum at the metal-semiconductor junction and rapidly decreases toward the bulk semiconductor. As the forward

bias increases to 0.8 V and 1 V, the peak electric field near the contact is significantly reduced and becomes more uniformly distributed across the device. This behavior reflects the reduction of the barrier height and the collapse of the depletion region, marking the transition from rectifying to quasi-Ohmic conduction at high bias.

3.3.4 Charge distribution

Similar to the Ohmic case, the electron concentrations are presented in Figure 20 and the hole concentrations in Appendix A:.

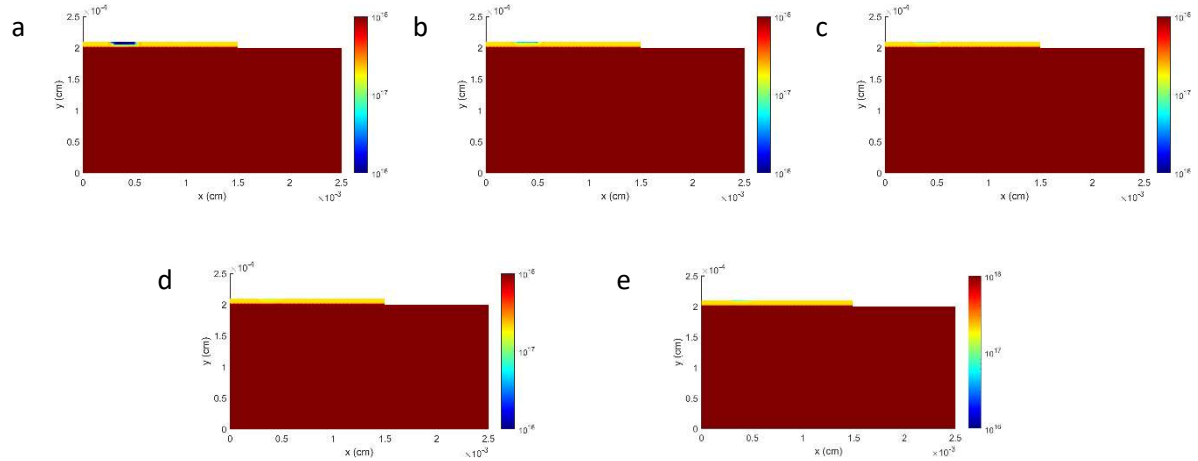
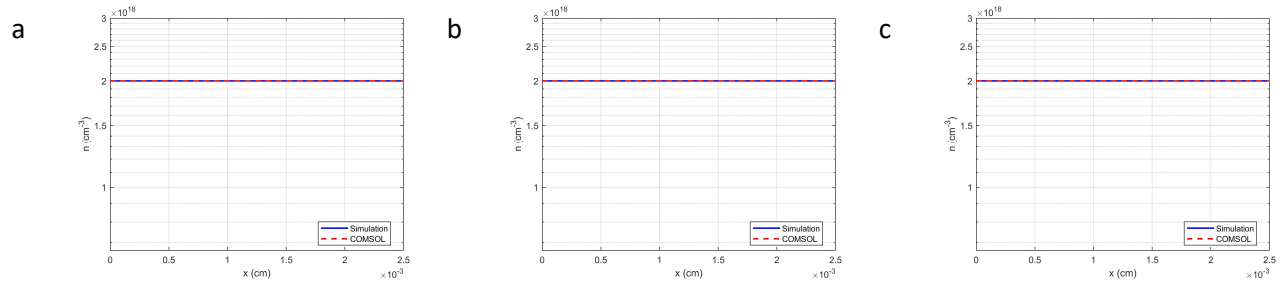


Figure 20. Electron distribution under a) 0V, b) 0.3V, c) 0.5V, d) 0.8V and e) 1V bias.

At 0 V, the electron concentration near the Schottky contact is markedly lower than in the bulk, since carriers are depleted in the barrier region. When the forward bias increases, the electron concentration at the interface rises sharply, indicating enhanced carrier injection and narrowing of the depletion layer. At 0.8V and 1 V, the carrier distribution becomes almost uniform across the semiconductor, confirming that the Schottky barrier is effectively flattened. This trend is consistent with the potential and electric-field results and represents the expected forward-bias behavior of a Schottky diode.

3.3.5 Cutline profiles

To provide a more quantitative view of the spatial variations under Schottky contact conditions, as before, the results at several horizontal and vertical cutlines were extracted, indicated in Figure 21-25 for the bias voltages of 0 V, 0.3 V and 0.8 V.



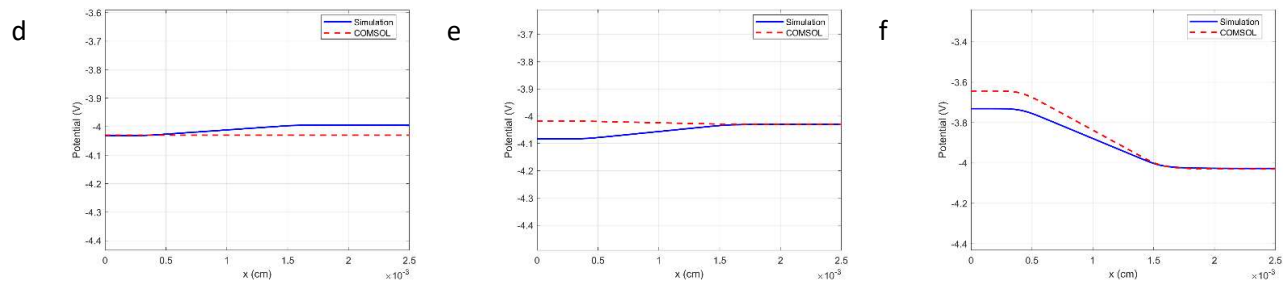


Figure 21. Cutline profiles of electron concentration and potential at $y = 1\ \mu m$ under Ohmic contact conditions. a) – c) are electron concentration distributions and d) – f) are potential distributions under 0V, 0.3V, 0.8V, respectively.

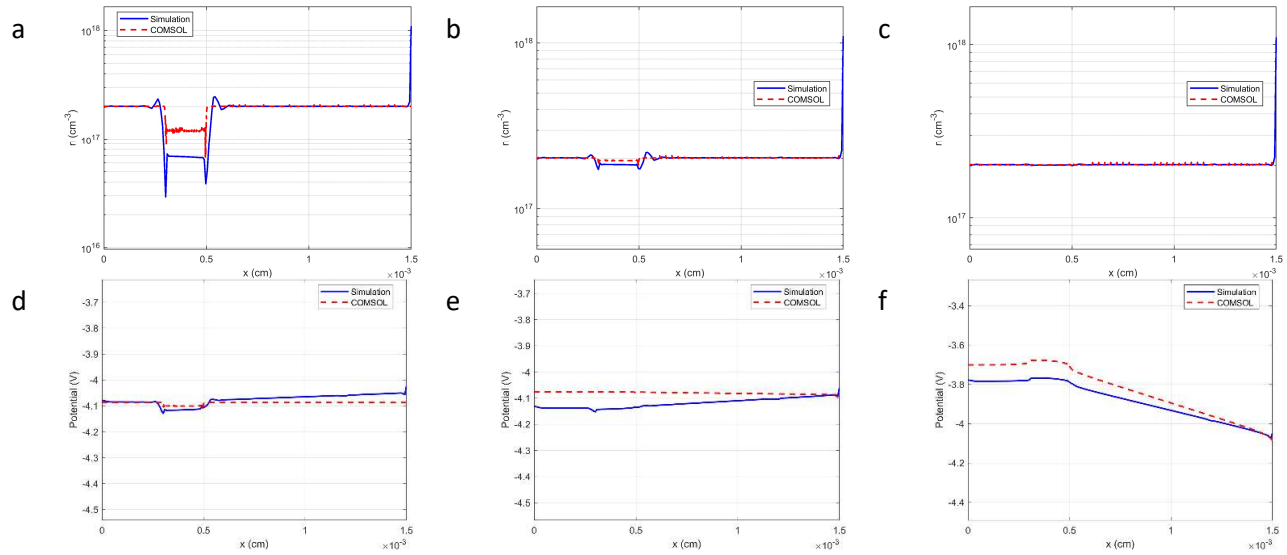


Figure 22. Cutline profiles of electron concentration and potential at $y = 2.05\mu m$ under Ohmic contact conditions. a) – c) are electron concentration distributions and d) – f) are potential distributions under 0V, 0.3V, 0.8V, respectively.

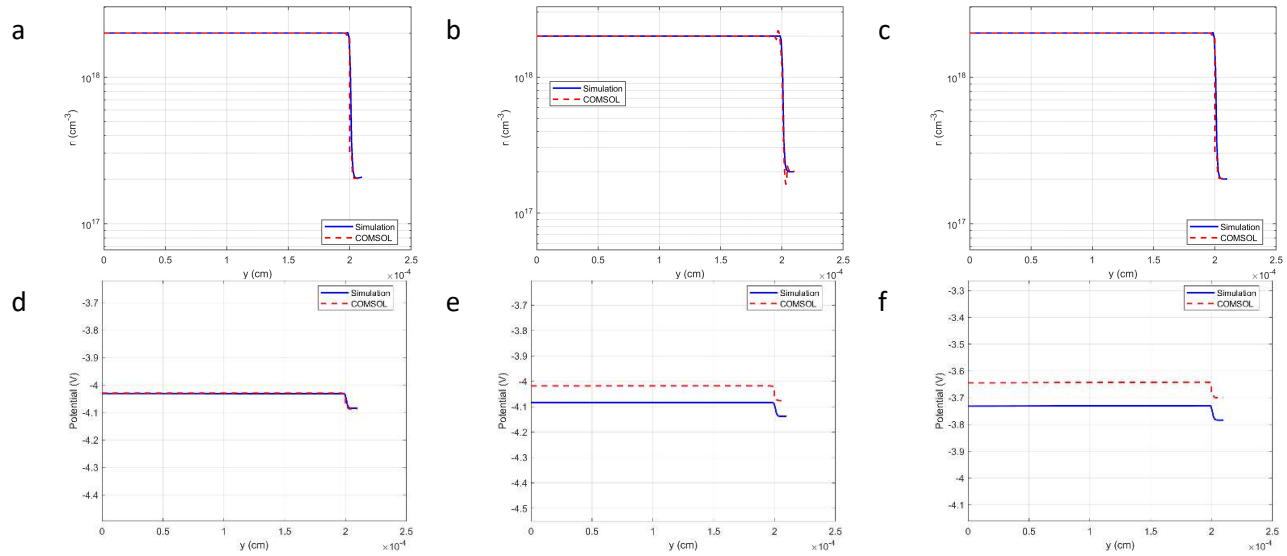


Figure 23. Cutline profiles of electron concentration and potential at $x = 2\ \mu m$ under Ohmic contact conditions. a) – c) are electron concentration distributions and d) – f) are potential distributions under 0V, 0.3V, 0.8V, respectively.

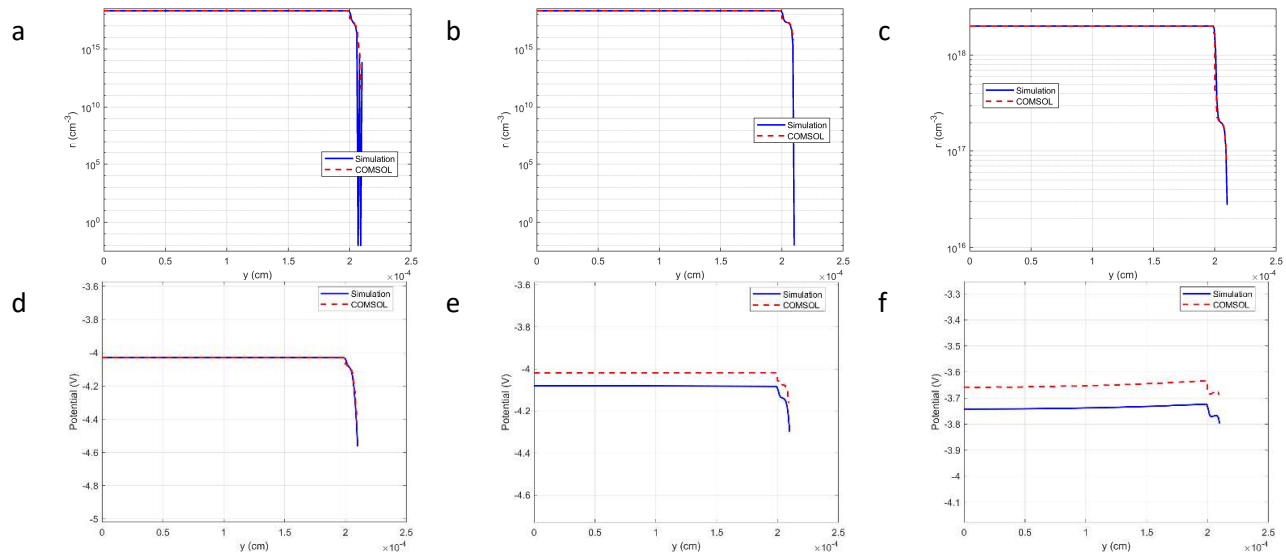


Figure 24. Cutline profiles of electron concentration and potential at $x = 4\ \mu m$ under Ohmic contact conditions. a) – c) are electron concentration distributions and d) – f) are potential distributions under 0V, 0.3V, 0.8V, respectively.

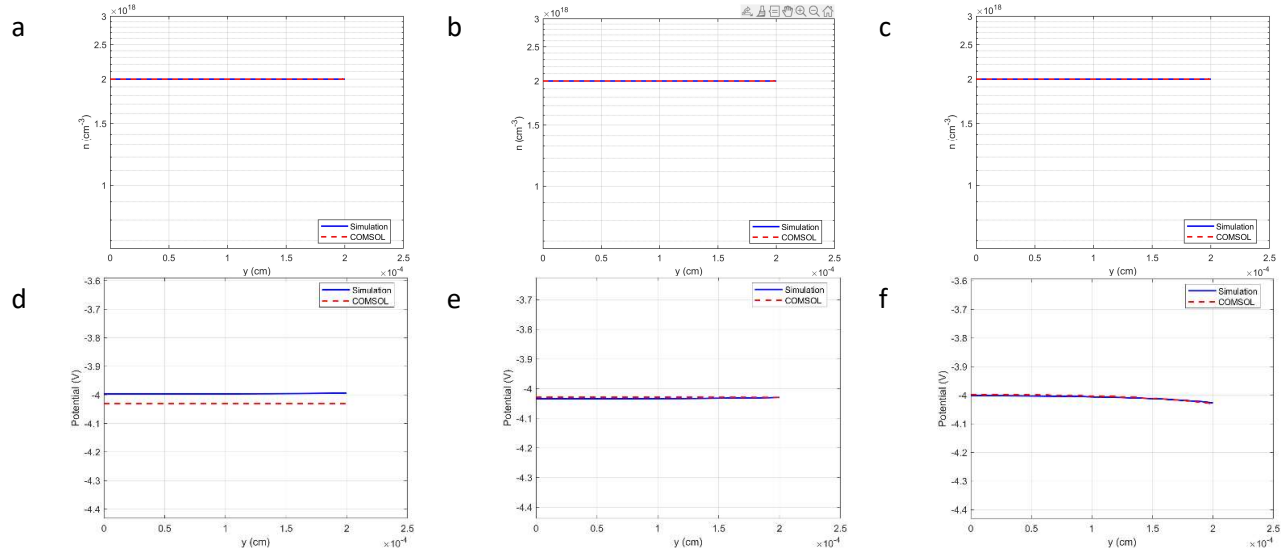


Figure 25. Cutline profiles of electron concentration and potential at $x = 15.2\ \mu m$ under Ohmic contact conditions. a) – c) are electron concentration distributions and d) – f) are potential distributions under 0V, 0.3V, 0.8V, respectively.

Along the horizontal direction, the electron concentration and potential distributions display clear deviations from the Ohmic case. When no bias is applied, the potential drop near the Schottky interface indicates ($y = 2.05\ \mu m$) the presence of the built-in potential barrier. As the forward bias increases to higher values, the potential drop becomes smaller, corresponding to a reduction in barrier height and narrowing of the depletion region. The same trend is observed for the electron concentration, which drops sharply under the electrode due to carrier depletion. In the n+ region, further away from the contact, the potential variation becomes smoother. This demonstrates that the depletion and barrier effects are strongly localized near the surface Schottky contact. The simulation and COMSOL results show very similar trends, although there exists discrepancy in the absolute values of the potential for all biases.

At $x = 2\ \mu m$, which lies just beside the metal-semiconductor junction, the potential begins to bend near the surface, and a small depletion of electrons is already visible. This indicates the influence of the Schottky barrier extending laterally beyond the physical edge of the electrode. At $x = 4\ \mu m$,

directly below the electrode, the potential exhibits a steep drop near the surface, corresponding to the Schottky barrier and the associated depletion region. The electron concentration is significantly reduced at the interface under no bias, which is considered to be a numerical artefact rather than a physically meaningful result. This discrepancy is likely related to the termination condition of the under-relaxation scheme or other aspects of the numerical convergence behavior. Then it gradually increases under forward bias, as the depletion layer narrows down and the barrier height decreases. At $x = 15.2\,\mu m$, far from the contact region, both the potential and carrier concentration distributions are nearly constant. This confirms that the electric field and depletion effects are strongly localized near the Schottky interface, while the bulk of the device remains quasi-neutral and Ohmic.

Notice that although the simulated results reproduce the physical trends well, small potential mismatches are observed compared with the COMSOL reference. These discrepancies mainly arise from the use of Dirichlet boundary conditions, which are fixed at a specific bias and therefore do not adjust dynamically with changes in applied voltage. In reality, quantities such as the Schottky barrier height and built-in potential vary non-linearly with applied bias due to thermionic emission and charge redistribution at the interface. As a result, the potential reference level effectively shifts with bias, leading to slight differences in absolute potential values across simulations. However, the relative potential drop and the carrier distribution trends remain I close agreement, indicating good consistency in the physical behavior reproduced by both models.

4 Summary

This thesis presented the implementation and validation of a physics-based DDM solver for planar GaAs Schottky diodes using the FEM. The work aimed at establishing a transparent, extendable, and fully numerical framework, capable of accurately describing carrier transport in semiconductor devices under different contact conditions.

The study first reviewed the theoretical background of metal-semiconductor contacts, highlighting the distinct physical mechanisms of the Ohmic and Schottky interfaces. Based on this, the coupled nonlinear PDE governing carrier transport and electrostatics were normalized and reformulated into a weak form suitable for the FEM implementation. The boundary conditions specific to Schottky and Ohmic contacts were developed and incorporated into the model. And numerical stability was ensured through a Gummel-type iteration with a successive under-relaxation method.

The implemented solver was verified against COMSOL for both Ohmic-Ohmic and Schottky-Ohmic configurations. Quantitative comparisons of potential, electric field and carrier density distributions showed excellent agreement between the developed model and the reference simulations, with only minor deviations. The $I-V$ characteristics under Schottky contact correctly reproduced the expected rectifying behavior, confirming the accurate implementation of the thermionic emission boundary conditions.

In summary, this thesis achieved the complete device-level modeling of GaAs Schottky diodes within the DDM-FEM framework, providing both physical accuracy and numerical flexibility. The developed code forms a foundation for future field-circuit co-simulation, where the device model can be coupled with external lumped circuit elements to simulate high-frequency nonlinear behavior. Ultimately, realizing such a coupled simulation environment would enable more reliable and efficient design of terahertz and microwave components based on Schottky diodes, bridging the gap between semiconductor device physics and circuit-level system modeling.

5 References

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Appendix A: Hole concentration distributions

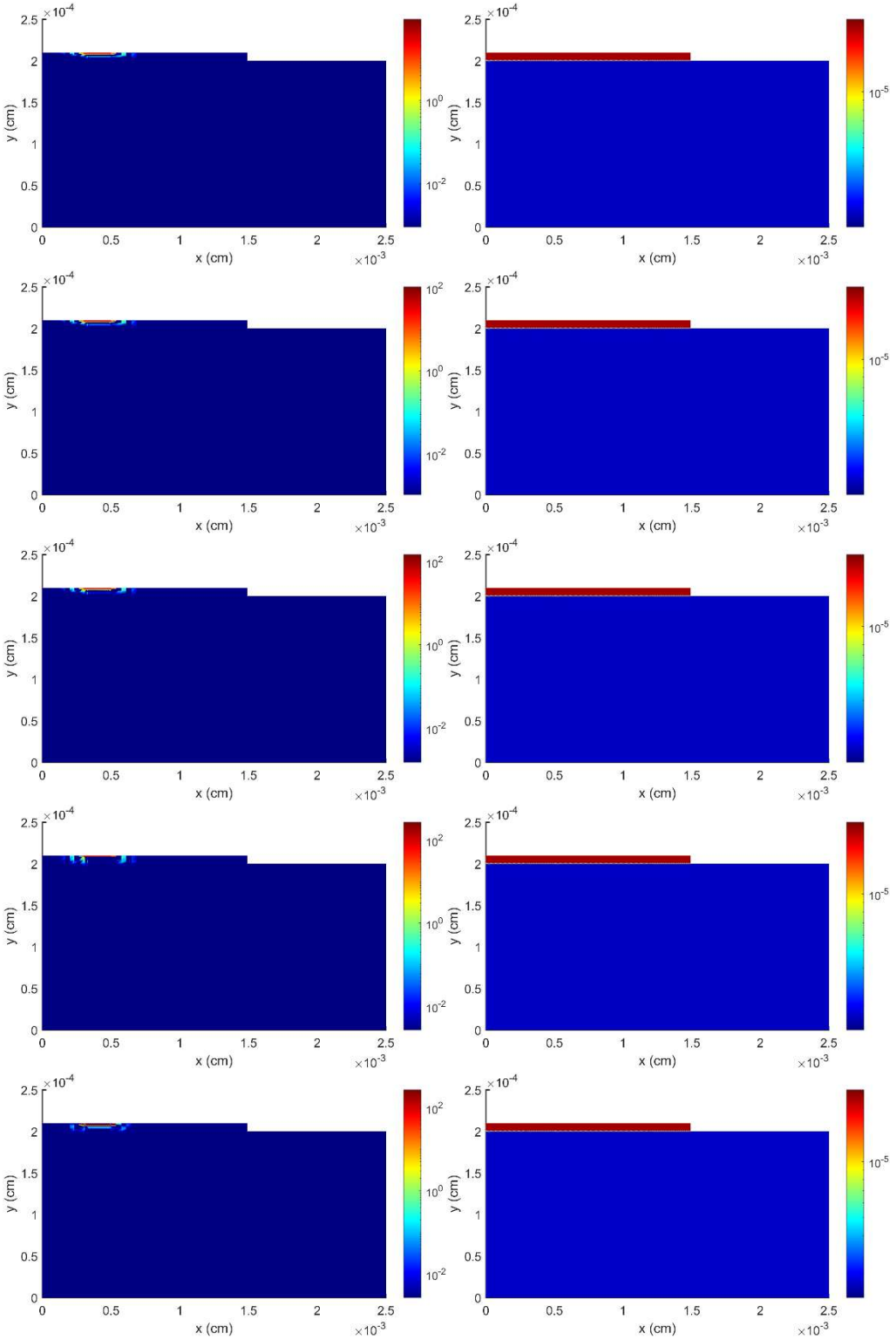
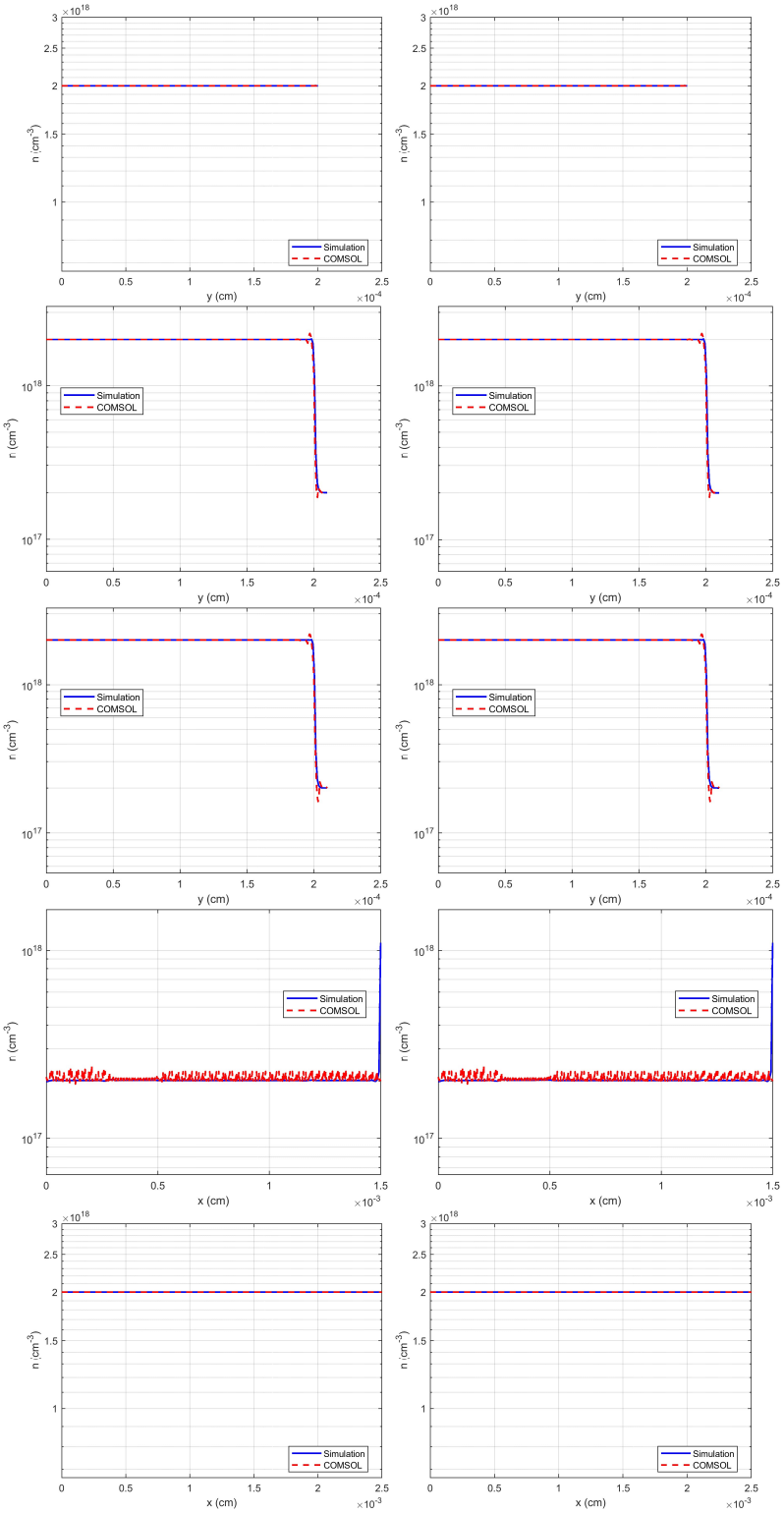


Figure 26. Hole concentrations under 0V, 0.3V, 0.5V, 0.8V and 1V respectively, left side represents Ohmic-Ohmic, and right one Schottky-Ohmic contacts.

Appendix B: Cutline profile comparisons



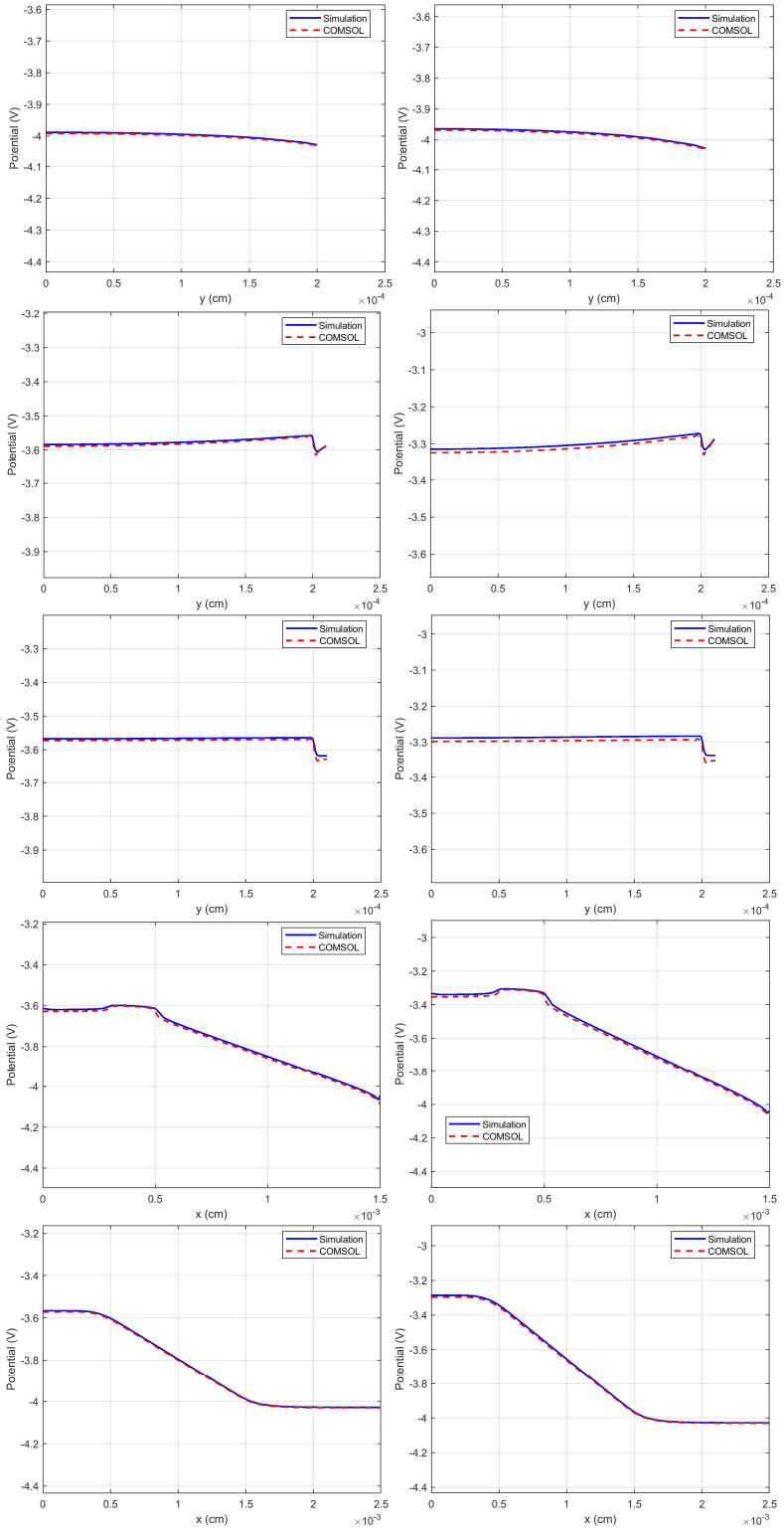
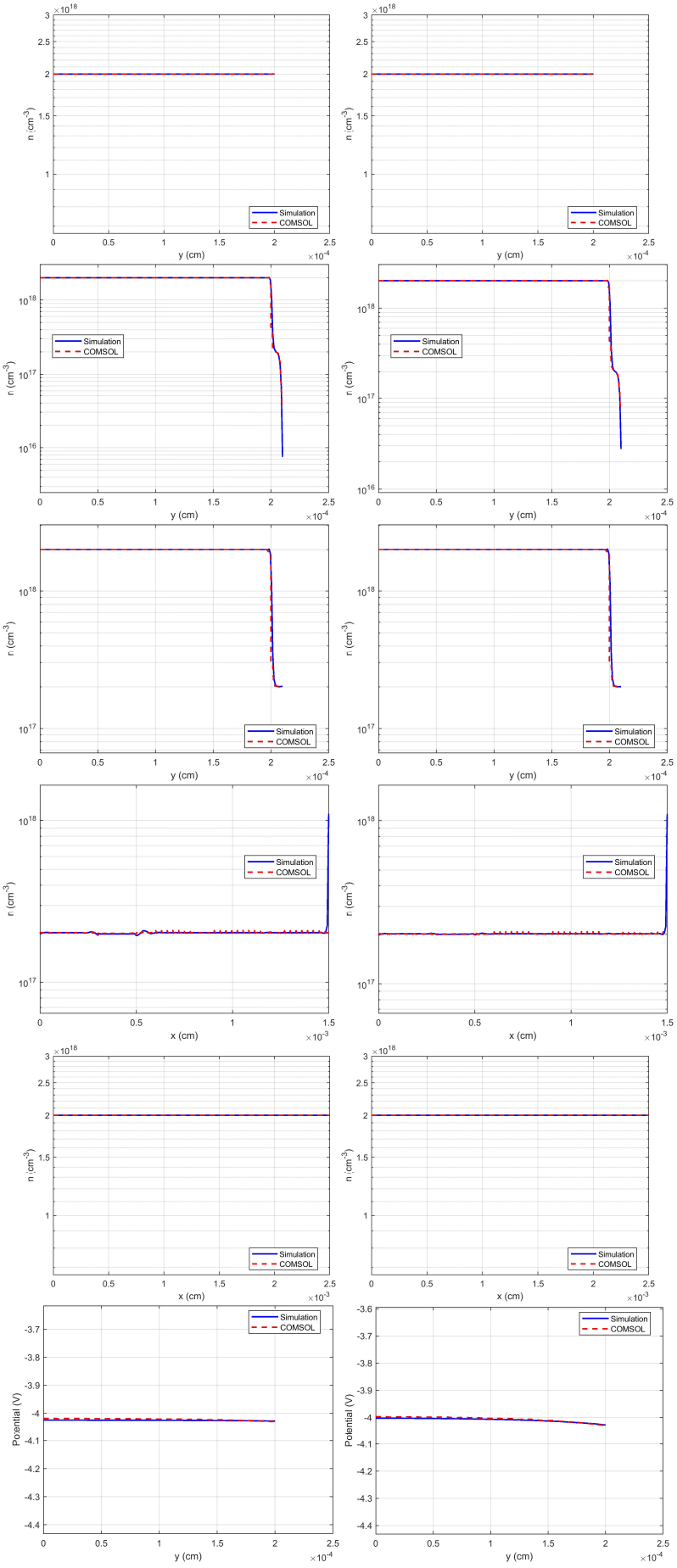


Figure 27. Cutline profiles under 0.5V(left) and 0.8V(right), with Ohmic-Ohmic contacts.

Cutline profile comparisons



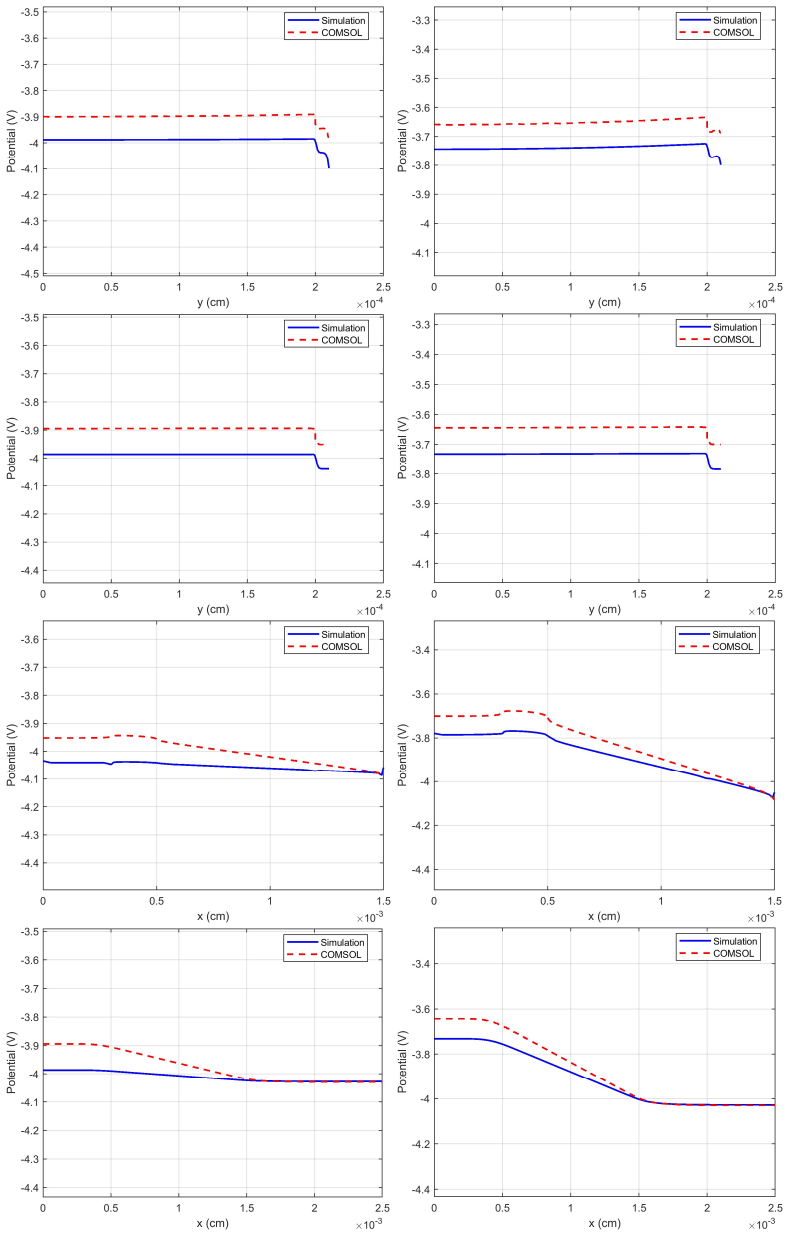


Figure 28 Cutline profiles under 0.5V(left) and 0.8V(right), with Schottky-Ohmic contact